An optimisation approach for supply chain procurement transport operational planning with 3D full-truckload constraints and returnable transport items

This paper proposes a mixed integer linear programming model, 3D-SCPTOP, for procurement transport planning in manufacturing supply chains, integrating the procurement, inventory and transport decisions of an assembly plant and its supplier with three-dimensional full-truckload constraints and the closed-loop management of returnable transport items. Because the placement subproblem is NP-hard, we reformulate it by Dantzig–Wolfe decomposition, taking the complete loaded truck as the unit of decision: every geometric, stacking, crush, payload and full-truckload constraint lives inside one pricing problem per direction, and a column keeps its layout, so a plan is executable by construction. The master is solved by branch and price over the fleet vector with a flow-first relax-and-fix construction as primal heuristic, and a certified bound of the volumetric relaxation is a valid lower bound for the 3D problem. On the volumetric problem the method returns a plan 0.0001% above a proved optimum. On the 3D problem, where the compact model returns no incumbent beyond five periods, it returns verified plans on all five instances of a ladder reaching twenty-four products and twenty periods, from 12 to 156 trucks within budgets of 900 to 3 600 s, with provably minimum fleets through twelve periods and certified gaps of 0.0075% to 0.163%, and of 14.5% and 10.7% where a per-stack crush limit binds. A volumetric model's plan prescribes loads proved impossible to place, so a volumetric transport capacity is not a conservative simplification; on the industrial catalogue the error runs one way, as the catalogue alone shows.

Keywords— Procurement, inventory, transport planning, full-truckload, 3D container loading

1 Introduction

Optimizing the use of resources in manufacturing supply chains is crucial for boosting economic growth, improving social well-being, and minimizing environmental impact. Integrating transport, inventory, and production processes is a fundamental characteristic of supply chains (Stadtler, Kilger, and Meyr 2015) because joint decisions will improve supply chain performance(Díaz-Madroñero, Mula, and Peidro 2017).

Procurement refers to the process of acquiring goods and services in a supply chain (Chopra and Meindl 2015). One of the main goals of procurement is to obtain raw materials (RM), parts or finished goods (FG) for a factory or distribution centre from a supplier. As mentioned in (Fleischmann, Kopfer, and Sürie 2015), the responsibility for transport planning usually lies with the supplier. However, sometimes the manufacturer makes transport decisions of its supplies in, for example, the automotive industry. Furthermore, an essential asset in the automotive supply chain is not considered in traditional material requirement planning (MRP): returnable containers or returnable transport items (RTI), which are used as tertiary packaging. RTIs enable the storage and transportation of RM and FG through the supply chain. Usually, the manufacturing plant supplies empty RTIs to its supplier, while the supplier delivers full RTIs with the FG or RM. Therefore, during the procurement process, transport and inventory decisions should be integrated, including RTI management. Here, we focus on optimising the operational decisions made during procurement transport planning.

According to (Díaz-Madroñero, Peidro, and Mula 2014), using algorithm-based methods for procurement transport planning is more effective than relying on manual approaches based on planners' intuition. Manual planning can be short-sighted and lead to less efficient outcomes. This becomes even more significant when optimizing procurement transport tasks that aim to reduce the number of trucks and inventory levels, especially under 3D loading constraints where the precise placement of containers within trucks must also be determined. Furthermore, the capacity constraints used in the scientific literature on procurement transport

planning are based mainly on freight volume or the total number of loaded product units (Hashemi-Amiri, Ghorbani, and Ji 2023) and (Díaz-Madroñero, Peidro, and Mula 2014). The current research seeks to address this gap by introducing a new mixed-integer linear programming (MILP) model. This model uniquely combines procurement transportation with considerations for 3D loading, inventory management, and RTI management. Hereafter, we refer to this model formulation as 3D supply chain procurement transport operational planning (3D-SCPTOP). Additionally, a branch-and-price procedure (BP-3D-SCPTOP) is developed to solve it.

In this paper, we present four main contributions, listed in the order of the evidence behind them. (i) We formulate a novel MILP model, 3D-SCPTOP, integrating procurement transport, RTI management and inventory decisions with stack-based full-truck loading constraints. (ii) We reformulate it by Dantzig–Wolfe decomposition over geometrically realisable truck loads, deriving the decomposition structurally — the model is written over trucks, so deleting the quantifier over (direction, truck, period) from its block rows is the pricing problem — and obtaining a master in which *no loading configuration appears at all*: the master couples to the geometry through two aggregate-flow definitions and one truck count per direction and period, and the pricing problem of a direction is the compact model's own block with the quantifier over trucks removed. We solve it by a branch and bound over the fleet vector with column generation over that mixed integer pricing problem and a relax-and-fix construction as primal heuristic, giving BP-3D-SCPTOP, and we show that a certified bound of the volumetric relaxation is a valid lower bound for the 3D problem. (iii) We establish the method's accuracy where a reference optimum exists — on the volumetric problem, against the compact formulation, which it matches to 0.0001% of a proved optimum (Section 8.2) — and then show that it keeps returning verified, executable plans and certified bounds in the 3D regime beyond which the compact model returns no incumbent at all. (iv) We show by placeability proof that a volumetric representation of truck capacity, as used in the procurement transport literature, yields plans that are cheaper on paper but cannot be loaded in practice, nor repaired by re-packing their own quantities.

Readers are referred to the typology by (Wäscher, Haußner, and Schumann 2007) for a global view of container loading problems, where the different dimensions of the loading problem are described. Furthermore, (Silva, Toffolo, and Wauters 2019) summarise the exact methods used in the literature to solve container loading problems, and (X. Zhao et al. 2016) explore other solution approaches in the literature. Readers are also referred to the literature reviews of (Mosca, Vidyarthi, and Satir 2019) and (Díaz-Madroñero, Peidro, and Mula 2015), which focus more on integrating inventory decisions and transport. Here, we jointly address container loading, procurement transport and inventory decisions.

The paper is organised as follows. Section 2 provides a literature review related to the problem under study. Section 3 presents the problem description. Section 4 describes the 3D-SCPTOP MILP formulation. Sections 5–7 develop the branch and price solution approach, BP-3D-SCPTOP, together with its pricing subproblem and bounding arguments. Section 8 reports the computational experiments. Finally, Section 9 concludes and puts forward further research lines.

2 Literature Review

The goal of this research is to integrate procurement transport operational planning decisions with 3D loading restrictions and RTI packaging. Based on the authors' research, no previous work has been identified that meets this criterion. Therefore, three independent, but sometimes overlapping, knowledge bodies have been reviewed: procurement transport and inventory planning, container loading problems and RTI literature.

2.1 Procurement transport and inventory planning

Models that integrate transport and inventory costs during the procurement process are widely studied in the literature. As noted in (Langley et al. 2020), transportation facilitates the movement of raw materials and finished products between suppliers, warehouses, manufacturers, and retailers. Therefore, transport processes play a crucial role in supply chains. (Fleischmann, Kopfer, and Sürie 2015) distinguish between two main types of transportation within a supply chain: "procurement transport," which involves moving materials from suppliers to manufacturing sites, and the "distribution of goods," which concerns delivering finished products from factories to customers. This article focuses specifically on addressing the challenges related to procurement transportation.

There are two main dimensions of procurement transport problems according to the supply chain configuration and the applied transport method. Regarding supply chain configurations (Mosca, Vidyarthi, and Satir 2019): (i) single source to a single destination (SS); (ii) multiple sources to a single destination (MS); (iii) single source to multiple destinations (SM); (iv) multiple sources to multiple destinations (MM).

Source components determine the number of supplier locations that are taken into account, while destination components denote the number of supplied manufacturers. Additionally, different transport methods are identified in (Díaz-Madroñero, Peidro, and Mula 2014):milk run, less-than-truckload (LTL), full-truckload (FTL) . We also analyze the formulation of the transportation capacity constraints in the scientific literature.

(Berman and Wang 2006) address the problem of optimising distribution strategies to deliver products from suppliers to plants, whose aim is to minimise transport, pipeline and plant inventory costs by using cross docks. A mixed integer non-linear programming (MINLP) formulation, a Lagrangian relaxation heuristic, a greedy heuristic, and a branch-and-bound algorithm are proposed to model and solve the problem.

(Sancak and Salman 2011) present a MILP model to optimise procurement transport decisions in a SS supply chain configuration using LTL and FTL transport methods. The objective is to minimise transport and inventory holding costs, while fulfilling production plan requirements on a multiperiod planning horizon. The study explores the possibility of delaying LTL shipments to the subsequent time period, and utilising safety stocks whenever necessary, in a stochastic environment by employing data from a bus manufacturer. The findings demonstrate that delaying shipments leads to reduced holding and transport costs with a minimal stockout risk.

(Schöneberg, Koberstein, and Suhl 2010) examine the concept of delivery profiles, which define the delivery frequency at a fixed rate, offering structured planning, minimizing management effort, and improving coordination in logistics. In this context, FTL transport is used when the goods' volume exceeds a vehicle's capacity, making direct shipment possible. LTL is applied when the volume is insufficient to fill a truck, and the goods are first sent to a consolidation center before reaching the final destination. In (Schöneberg, Koberstein, and Suhl 2013), the proposed model accounts for demand uncertainty.

(Díaz-Madroñero, Peidro, and Mula 2014) present a fuzzy multi-objective integer linear programming (FMOILP) model that focuses on operational procurement transport planning in an automotive supply chain, which involves a single source and a single destination while utilising FTL. The developed model aims to minimise the number of trucks employed for procurement transport and the inventory level by considering the uncertainties related to transport capacity levels.

(Hein and Almeder 2016) investigate the benefits of integrated procurement transport and production planning for supply and production processes by comparing a sequential approach, where production is firstly determined, followed by procurement transport decisions, plus an integrated approach in which decisions are jointly made. Numerical experiments reveal that the integrated planning approach leads to substantial cost savings.

(Díaz-Madroñero, Mula, and Peidro 2017) present an novel MILP model that integrates procurement, production and transport planning decisions within manufacturing systems. This model extends beyond traditional material requirement planning (MRP) systems by incorporating procurement transport decisions from multiple sources and a single destination into a unified optimization framework. It considers the capacities of both material and production resources, along with procurement transport planning decisions, by employing various shipping modes such as FTL, LTL, and milk-run.

(Chitsaz, Cordeau, and Jans 2019) address integrating production, inventory and procurement transport into a manufacturing context by defining this problem as an assembly routing problem (ARP). It introduces a novel MILP model for the ARP and proposes a three-phase decomposition matheuristic for its solution.

(Baller et al. 2022) examine the optimization of procurement transport in an automotive factory, focusing on first-tier suppliers at an aggregate level across multiple sources (570) to a single destination, using FTL, LTL, and courier express service (CES) as transportation methods. The proposed MILP approach includes a cyclic inventory constraint, truck utilization, and storage limitations.

(Hashemi-Amiri, Ghorbani, and Ji 2023) propose an integrated approach to optimize a perishable food supply chain. The model combines supplier selection, procurement and distribution transport, production scheduling, and vehicle routing to improve supply chain efficiency and resilience. It handles demand uncertainties through a distributionally robust chance-constrained model. The model is reformulated as MILP by utilizing duality and linearization techniques.

2.2 Procurement Transport with RTIs

Some models further integrate the procurement and delivery of RTIs into the procurement transport literature. In this sense, RTIs are procured to the supplier, who loads the items and sends them to customer. We extend the supply chain definitions previously defined to contemplate the RTIs management strategies.

(Soysal et al. 2023) develop a tactical-level, multi-period closed-loop Vendor-Managed Inventory (VMI) system for RTIs in an agro-food supply chain. The SM supply chain integrates forward flows of full RTIs from a single vendor to multiple customers and reverse flows of empties back to the vendor for cleaning . All

routing and inventory decisions are made centrally by the vendor under full information sharing. Sustainable plans in terms of amounts of energy spent and empty crates collected are devised and evaluated.

(Guo et al. 2024) model two-tier Container Management Centre (CMC) network . Empty RTIs are dispatched from central CMC hubs to upstream suppliers, loaded with products, carried by truck to downstream customers, then returned empty to regional CMCs for cleaning and storage before being shipped back to central hubs. A neural network is used for demand forecasting. A MILP is proposed that maximises total profit by optimally determining transport costs and the penalties applied to late RTIs returns.

(Y. Zhang et al. 2024) formulate a perishable-food inventory-routing MILP with multiple RTI types in a SM supply chain, integrating procurement of products and pickup of both filled and empty RTIs. Decisions are made on a operational level, optimally determining per period the quantities to pack, deliver, pick up, purchase and lose, as well as the vehicle routes in order to maximise the total profit of the supply chain that is equal to the total quality-dependent revenue minus the total costs involved, including inventory holding costs, product cost, transportation costs, and RTI purchasing and loss costs. The model is solved exactly via CPLEX solver and approximately via a tailored Kernel Search matheuristic.

2.3 Container loading

Container loading problems encompass a subset of geometric assignment problems known as cutting and packing ($C\&P$) problems. They revolve around the allocation of n-dimensional orthotope items in larger n-dimensional orthotope containers. Here the aim is to optimise an objective function while taking into account two key geometric constraints: complete containment of items inside containers and lack of item overlap. A loading pattern serves as a formal representation of a solution to such an assignment problem. Dyckhoff introduced in (Dyckhoff 1990) a typology that unifies $C\&P$ problems using four different criteria: dimensionality of space, the objective of assignment, containers types and box type. Consequently, (Wäscher, Haußner, and Schumann 2007) develop a new typology that has become the standard in the $C\&P$ literature. We briefly describe the typology used in (Wäscher, Haußner, and Schumann 2007) based on five different criteria: (i) dimensionality of space; (ii) kind of assignment (input minimisation of containers for all small items and loading patterns; output maximisation to pack a subset of small items with limited container availability;) (iii) assortment of small item by distinguishing if they are either a weakly (or strongly) heterogeneous assortment or identical small items ; (iv) assortment of large objects (if it is either several large objects or one large object); (v) shape of small items (circular or of another shape). In that typology's names, the problems cited below are the single large object placement problem (SLOPP), the multiple identical large object placement problem (MILOPP) and the single knapsack problem (SKP) on the output-maximisation side, and the single stock size cutting stock problem (SSSCSP), the multiple stock size cutting stock problem (MSSCSP) and the single and multiple bin size bin packing problems (SBSBPP, MBSBPP) on the input-minimisation side; 3D-BPP is the three-dimensional bin packing problem.

The first mathematical programming model in a scientific journal to describe the container-loading problem was developed in (C. S. Chen, Lee, and Shen 1995), where the authors introduced an MILP model to address the 3D input minimisation problem by considering rotations and containers of varying dimensions. However, it does not specify the use of weakly or strongly heterogeneous boxes or containers. Additionally, real-world constraints are modelled like, for example, weight distribution.

(Junqueira, Morabito, and Sato Yamashita 2012b) propose an alternative MILP formulation to (C. S. Chen, Lee, and Shen 1995) using binary encoding for box position, which utilises normal patterns to reduce possible box coordinates by considering multiples of boxes dimensions and ensuring optimal solutions. Additionally, loading and cargo constraints are developed to ensure the static and dynamic stability of the cargo. Boxes must be supported by other box top faces or the container floor. The same authors integrate further loading restrictions in (Junqueira, Morabito, and Sato Yamashita 2012a), where they integrate multidrop constraints, which address the challenge of packing a truck with boxes for different customers at various destinations. The goal is to plan the container's loading to minimise the time spent unloading and reloading boxes at each destination.

(Zhu and Lim 2012) devise a tree search-based method to solve SLOPP and SKP with 3D constraints. Their approach involves evaluating free spaces by employing a look-ahead tree search and selecting optimal placements based on the best free space-block pairs.

(Gonçalves and Resende 2013) develop a biased random-key genetic algorithm (BRKGA) designed to solve SBSBPP. The algorithm addresses geometric constraints and introduces a maximal-space representation to handle available free spaces in bins. A novel placement procedure is hybridised with the BRKGA, which optimises the order of box packing and the parameters utilised by the placement procedure to offer an effective solution approach to an NP hard problem.

(Bortfeldt and Wäscher 2013) identify constraints for $C\&P$, including item-related constraints (orientation constraints, load bearing, loading priorities), container-related constraints (weight distribution and weight limit), and cargo-related constraints (complete-shipment constraints) and load-related constraints (complexity constraints or stability constraints). The study also highlights the achievements and deficits in container loading research that pertain to these constraints.

(Paquay, Schyns, and Limbourg 2016) propose an MILP model for the MBSBPP. One noteworthy constraint is the assumption that containers are truncated parallelepiped instead of rectangular boxes. They also consider constraints, such as box rotation, container load bearing, weight distribution, stability and fragility of items.

(Mahvash, Awasthi, and Chauhan 2018) introduce a novel resolution procedure for the SBSBPP through the column generation technique, where the columns that are generated are the loading configurations of a single bin. For the pricing develop problem they develop a heuristic to find new columns with a negative reduced cost. This approach not only produces better results than the previous articles on the literature, but also gives new lower bounds.

(Silva, Toffolo, and Wauters 2019) study SLOPP models with 3D constraints. These authors compare the different approaches that exist in the literature for model packing problems based on two fundamental approaches: discretised formulation (Junqueira, Oliveira, et al. 2013) and continuous formulation (C. S. Chen, Lee, and Shen 1995) with subsequent modifications. They also indicate that discretised formulation(Junqueira, Morabito, and Sato Yamashita 2012a) performs the best for instances with similar sized boxes. However, it encounters memory limitations for large instances. For other sets of instances, the adapted formulations of (C. S. Chen, Lee, and Shen 1995) and (Paquay, Schyns, and Limbourg 2016) outperform other adaptations. Further research topics suggest including the design of new formulations by considering the limited approaches in the literature.

(Kurpel et al. 2020) introduce techniques and exact approaches to solve input minimisation and output maximisation problems for container-loading problems. Their methods consider practical constraints, such as orientation, separation of boxes and vertical stability. A set partitioning method is also employed in order to obtain upper bounds in a container-loading input minimisation problems. The authors adapt and evaluate four discretisation methods found in the existing literature and incorporate mathematical formulations to address the aforementioned constraints.

(Alonso, Alvarez-Valdes, and Parreño 2020) develop an algorithm by the greedy randomised adaptive search procedure (GRASP) to tackle a practical container-loading problem with time-sensitive constraints. Their approach addresses dynamic stability constraints while stacking products on pallets and loading them onto trucks during different time periods. Their approach provides high-quality solutions and surpasses an integer programming formulation in efficiency and effectiveness terms.

(Ali et al. 2022) conduct a comprehensive literature review by identifying constraints and solution methods for offline problems with full knowledge of the items to be packed and those tackling the online problems, where items are packed immediately when they arrive, and no prior knowledge of an item is known. The study also identifies and classifies solution approaches for offline packing into six categories: exact methods, constructive heuristics; local search and metaheuristic methods; approximation algorithms; machine learning and hyper-heuristic methods; tree search methods. (Harrath 2022) proposed a three stage heuristic based on layer generation to find good quality solutions the SBSBPP problem for large instance up to 993 bins and 200000 items taking into account load balancing constraints.

(Tresca et al. 2022) develop a matheuristic to solve the SBSBPP that includes items with different IDs and different categories, to be delivered to different clients. The algorithm creates monoitem, monocategory, and mixed layers which then are used to compute the final loading configurations of the bins.

(Li, M. Chen, and Huo 2022) develop a hybrid adaptive large neighborhood search algorithm to solve the multiple container loading problem, which is tries pack a given set of items with the minimum number of containers while maximizing the space utilization of them. Therefore, it is both a MBSBPP and a MSSCSP. Their approach takes into account some practical applications such as suspension constraints, rotation and weight limit.

(M. Chen, Huo, and Duan 2023) present a new approach to the 3D-BPP, introducing the 3D-BSDPP which focuses on optimizing both the design of bin sizes and the packing process. A novel Hybrid Biogeography-Based Optimization (HBBO) algorithm is proposed, integrating differential evolution and a local search strategy for enhanced performance.

One noteworthy trend in $C\&P$ is integrating loading problems with routing. This is known as capacitated vehicle-routing problem with 3D or 2D loading constraints (3L-CVRP or 2L-CVRP). (Junqueira, Morabito, and Sato Yamashita 2012b) present a MILP model with binary encodings for 3L-CVRP by including loading, cargo and multidrop constraints. The combination of these two NP-hard problems requires using heuristics and metaheuristics for real-world instances, as seen in (Dominguez et al. 2016), (Alonso, Martinez-Sykora,

et al. 2022) and (Wei et al. 2015). Another noteworthy trend is to use reinforcement learning techniques to solve $C\&P$, as in (Que, Yang, and D. Zhang 2023) and (H. Zhao et al. 2022).

2.4 Research Gaps & Methodology

Here, we discuss the main differences between the 3D-SCPTOP model proposal and those reviewed by specifically focusing on procurement transport. The key distinction of the 3D-SCPTOP model compared to the existing literature is the incorporation of 3D loading constraints, with vertical stability ensured by construction (Section 3), to model transport capacity with a view to providing more realistic solutions and reducing the likelihood of infeasible outcomes. Table 1 presents a comparison of inventory modelling, transport modelling, and supply chain configuration between the 3D-SCPTOP model proposal and those currently in the literature. Hence, the 3D-SCPTOP approach is based on a SS supply chain configuration, and takes into account the FTL transportation mode.

Article (Guo et al. 2024)	Supply Chain Configuration MM	Decision level TO	Transportation Mode FTL	Transportation Capacity Constraint UH	Transportation Cost X	Inventory Capacity Restriction X	RTIs S—NF	Inventory Costs X	Safety Stocks	Modelling Framework MINLP
(Berman and Wang 2006)	MM	T	FTL—LTL	V	X			X		MINLP
(Sancak and Salman 2011)		O	LTL—FTL	V	X			X	X	MILP
(Schöneberg, Koberstein, and Suhl 2010)	MS	T	FTL—LTL	V	X	X		X		MILP
(Schöneberg, Koberstein, and Suhl 2013)	MS	O	FTL—LTL	V	X	X		X		MILP
(Díaz-Madroñero, Peidro, and Mula 2014)	SS	O	FTL	UH	X			X	X	FMOILP
(Hein and Almeder 2016)	MS	O	MR	V	X			X		MILP
(Díaz-Madroñero, Mula, and Peidro 2017)	MS	T	FTL—LTL—MR	V	X	X		X	X	MILP
(Meyer and Amberg 2018)	MS	T—O	MR—FTL	V	X	X		X		MILP
(Chitsaz, Cordeau, and Jans 2019)	MS	O—T	MR	V	X	X		X		MILP
(Baller et al. 2022)	MS	O	MR—FTL—LTL—CES	V—W	X	X			X	MILP
(Hashemi-Amiri, Ghorbani, and Ji 2023)	MS	T	MR	V	X					CCP
3D-SCPTOP	SS	O	FTL	3D-Constraints	X	X	M—F	X	X	MILP
(Y. Zhang et al. 2024)	MS	TO	MR	UH	X	X	M—NF	X		ILP

Abbreviations. **T** (tactical), **O** Operational; **FTL** (full-truck load), **LTL** (less-than-truck load), **MR** (milk-run), **CES** (courier express service); **V** (volumetric), **W** (weight), **UH** Unit Holding; **CCP** chance-constrained programming; **TO** (tactical and operational); **S**, **M** (single, multiple RTI types), **NF**, **F** (non-foldable, foldable RTIs); **ILP** (integer linear programming)

Table 1: Comparison of procurement transport models

On the other hand, the 3D-SCPTOP model diverges from those proposed in the packaging literature in several aspects. Firstly, the loading problem takes into account different time periods. Therefore, the specific container (small items or boxes in the $C\&P$ literature) that we utilise are yet to be defined, as the choice of container depends on the inventory levels of each product. Also, the objective function depends on inventory and transport costs, which are inherently connected to the loading problem at hand. Therefore, we simultaneously solve an input minimisation and output maximisation problem. Both of these are a result of transport minimisation and inventory costs. Input minimisation comes because we want to find the minimum number of trucks needed. Meanwhile, output maximization comes from the FTL and the transport costs.

Based on the literature review and the identified research gaps, our proposal is structured according to the steps depicted in Figure 1.

Article	Problem type	Resolution procedure	Orientation	Weight	Load Balance	Stability	Time-Dependent
(Zhu and Lim 2012)	SLOPP—SKP	TSM	X	X			
(Gonçalves and Resende 2013)	SBSBPP	BRKGA	X	X		X	
(Paquay, Schyns, and Limbourg 2016)	SBSBPP	EM	X	X	X	X	
(Mahvash, Awasthi, and Chauhan 2018)	SBSBPP	CG	X	X			
(Alonso, Alvarez-Valdes, and Parreño 2020)	SSSCSP	GRASP	X	X	X	X	X
(Tresca et al. 2022)	SBSBPP	MH	X	X	X	X	
(Harrath 2022)	SBSBPP	H	X	X	X		
(Li, M. Chen, and Huo 2022)	SBSBPP—MSSCSP	HALNs	X	X		X	
(M. Chen, Huo, and Duan 2023)	BSDPP	HBBO	X	X			
3D-SCPTOP	SSSCSP & MILOPP	BP	X	X		X	X

Abbreviations. **T**SM (Tree search method), **BRKGA** Biased Random-Key Genetic Algorithm; **EM** (Exact Methods), **MH** (MatHeuristic), **CG** (Column Generation), **H** (Heuristic); **BP** (Branch and Price); **HALNs** (Hybrid Adaptive Large Neighborhood Search), **HBBO** (Hybrid Biogeography-Based Optimization); **BSDPP** (bin size design and packing problem)

Table 2: Comparison of C&P problems

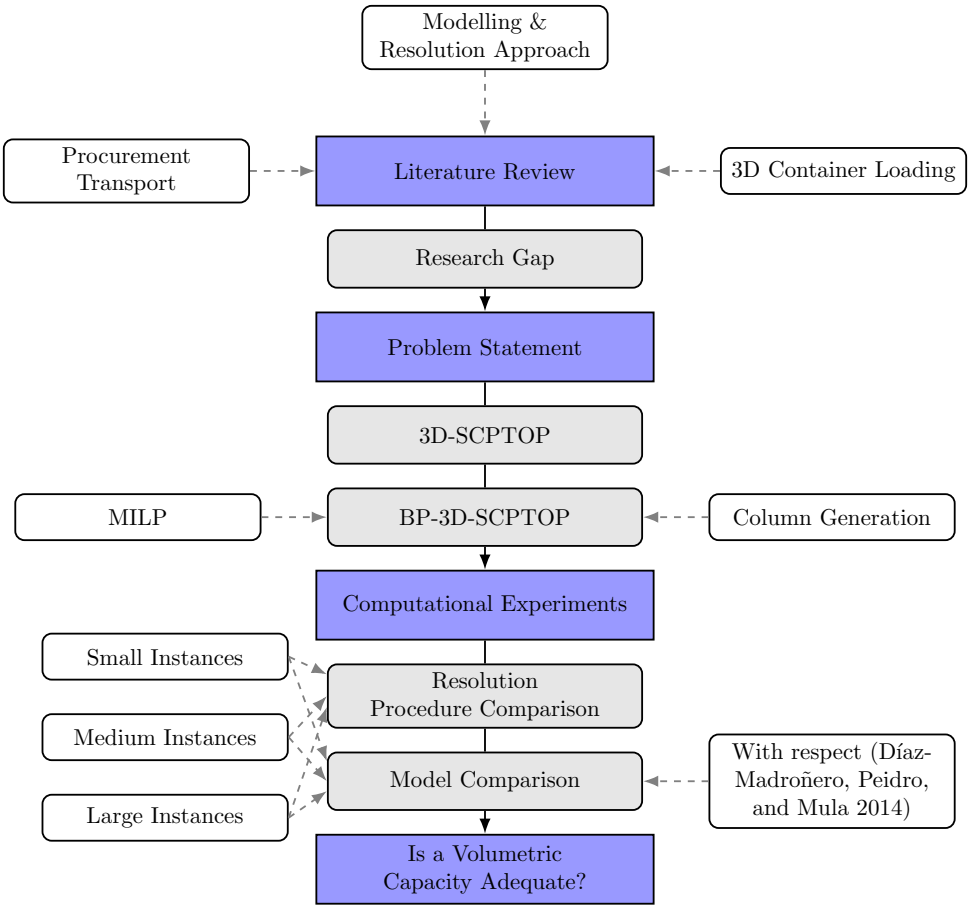


Figure 1: Research Methodology

3 Problem description

A plant assembles products supplied by a single supplier over a finite horizon. Every product travels in exactly one type of RTI: a rigid container, *erected* while it carries product and *folded* when empty, so one physical container occupies two different volumes depending on which way it is moving. The supplier packs its production into erected RTIs and ships them to the plant; the plant consumes the product, folds the containers and returns them. Both legs are run with identical trailers, every dispatch costs the same whatever it carries, and a trailer may be dispatched only if it is full. A dispatch arrives after a lead time of $lt \geq 1$ periods and is available at neither end in between, and each node must end every period holding enough stock to cover the next periods of its own activity. What has to be decided is, per direction and period, how many trucks go, what each carries and what consequently sits in stock; the cost is the number of dispatches plus the inventory held.

Two features separate this from a transport-and-inventory model. The two flows are coupled through the containers themselves — the plant can return only empties it has emptied and the supplier can fill only containers it has received — so a shortage on the return leg propagates into the delivery leg one lead time later. And the admissibility of a truck load is a three-dimensional packing condition rather than a capacity inequality: a load whose volume and weight both fit may still have no arrangement inside the trailer. Loading is represented through *stacks* of whole homogeneous layers rather than through the placement of individual RTIs, which makes the stacks self-supporting, so no vertical-stability constraint is needed.

4 The 3D-SCPTOP formulation

Throughout, $g \in G = \{s, p\}$ indexes nodes and directions (named by destination), $n \in N$ products, $r \in R$ RTI types, $h \in H$ stack types, $i, j \in I$ slot positions, $k \in K$ trucks and $t \in T$ periods, with $\rho(n)$ the RTI

type product n travels in and $M(r) = \{n : \rho(n) = r\}$; variables whose direction is fixed by definition omit g. Table 3 is the nomenclature.

The model is written over *trucks*. A dispatched truck designs its own layout — the stack types, orientations, positions and layer compositions of its slots are variables of that truck and that period, and its load is assigned to those slots in the same period — so there is no catalogue of loading configurations. That is the decision Section 5 rests on: every row below carries either the triple (g, k, t) or none of it, so the model is block-angular over $G \times K \times T$ and the pricing problem of the Dantzig–Wolfe reformulation is obtained by deleting that quantifier and nothing else.

Table 3: Nomenclature. The pricing problem of Section 6 is one block of this model with the index (g, k, t) deleted, so it reuses these symbols with that index dropped; the three variables it adds are listed last.

Indices and sets	
G	Supplier and plant nodes, $G = \{s, p\}$; $g = s$ is the folded direction (plant to supplier), $g = p$ the full one.
N, R, H	Products, RTI types, stack types.
I	Stack-slot indices on a truck floor, bounded by (46).
K, T	Trucks available per direction and period; planning periods.
$\rho(n)$, $M(r)$	The RTI type product n travels in; the products type r carries, $M(r) = \{n : \rho(n) = r\}$.
$R(h)$	RTI types compatible with stack type h.
Parameters	
cK	Cost of dispatching one truck.
lt, SC	Lead time, $lt \geq 1$; coverage window, in periods.
i_{gn}, ie_{gr}	Initial inventory at g of full RTIs of n, of folded RTIs of r.
ic_{gn}	Inventory holding cost of n at g.
d_{gnt}	Production plan for n at g in t: filled at s, consumed at p.
a_{nt}, ae_{rt}	Pipeline arrivals at p, respectively s, in $t \leq lt$.
ss_{gnt}, sse_{gtr}	Coverage stock of full, respectively folded, RTIs (20).
w_n, w^{f}_r	Weight of a full RTI carrying n; of a folded RTI of type r. Both > 0.
pK^w, $W^{\max}$	Truck payload; per-stack crush-limit cap on the *total* weight of one stack, over all its layers.
pK, qK, rK	Truck floor length, floor width, inner height; $\Gamma_K = pK \cdot qK \cdot rK$.
α	Minimum truck utilisation, on whichever of volume or payload binds, $0 < \alpha \leq 1$.
p_h, q_h	Footprint length and width of stack type h.
r_{gr}, a_r, γ_{gr}	Height of one RTI of type r at node g (full at p, folded at s); its footprint area; its volume $\gamma_{gr} = a_r r_{gr}$.
n_{hr}	RTIs of type $r \in R(h)$ per layer of stack type h; one layer tiles the footprint, $n_{hr} a_r = p_h q_h$.
$\hat{w}_{gr}$	Weight per RTI in the layer-capacity form of the utilisation floor: $\hat{w}_{sr} = w^{\mathrm{f}}_r$, $\hat{w}_{pr} = \max_{n \in M(r)} w_n$.
η^K, η^I	Weights of the transport and inventory objectives; both 1 throughout.
M^x, M^y	Big-M constants of the non-overlap rows, $M^x = pK$, $M^y = qK$.
$\bar{u}^V_n$, $\bar{u}^{\mathrm{e}}_r$	$\lfloor W^{\max}/w_n \rfloor$ full RTIs of n per slot; $\lfloor pK^w/w^{\mathrm{f}}_r \rfloor$ folded RTIs of r per truck.
Decision variables — the truck and its layout	
Z_{gkt}	$\in \{0, 1\}$. Truck k of direction g in t is configured, i.e. carries a layout.
Y_{gkt}	$\in \{0, 1\}$. That truck is dispatched.
θ_{gkt}	$\in \{0, 1\}$. Selects the resource on which its utilisation floor is required: volume if 1, payload if 0.
X_{gihkt}	$\in \{0, 1\}$. Slot i holds a stack of type h.
L_{gihktr}	$\in \mathbb{Z}_{\geq 0}$. Layers of RTI type $r \in R(h)$ in slot i.
LX_{gihkt}, LY_{gihkt}	$\in \{0, 1\}$. Stack h's length lies along the x-, respectively y-axis in slot i.
PX_{gikt}, PY_{gikt}	≥ 0. Bottom-left coordinates of slot i on the floor.
A, B, C, D (on $gijkt$)	$\in \{0, 1\}$. Which of the four separating half-planes resolves the overlap of slots i and j.
Decision variables — load and inventory	
V_{kint}	$\in \mathbb{Z}_{\geq 0}$. Full RTIs of n in slot i of plant-bound truck k in t.
V^{e}_{krt}	$\in \mathbb{Z}_{\geq 0}$. Folded RTIs of type r on supplier-bound truck k in t.

I_{gnt}, Ie_{gtr}	$\in \mathbb{Z}_{\geq 0}$. End-of-period inventory of full RTIs of n, of folded RTIs of r, at g.
Derived quantities (abbreviations, (1)–(5))	
$\overline{V}_{nt}$, $\overline{V^{\mathrm{e}}}_{rt}$	Full RTIs of n delivered to the plant in t; folded RTIs of r returned to the supplier.
$\overline{V}_{knt}$	Full RTIs of n on plant-bound truck k in t.
$\widehat{X}_{gikt}$, $\overline{L}_{ghktr}$	Slot i is used; layers of type r under stack type h on that truck, over all slots.
Added by the reformulation (Sections 5.2–6)	
$\lambda^g_{\omega t}$	$\in \mathbb{Z}_{\geq 0}$. Trucks of class (g,t) dispatched carrying column ω.
$\pi_{nt}, \pi_{rt}, \mu_{gt}$	Duals of (55)–(56), respectively of (57) and of every further row a dispatch variable of (g,t) occupies.
n_{gt}, X	Dispatch total of class (g,t), $n_{gt} = \sum_\omega \lambda^g_{\omega t}$; the fleet $X = \sum_{g,t} n_{gt}$.

4.1 Derived quantities

Five sums recur, as *abbreviations* and not constraints.

$$\overline{V}_{nt} = \sum_{k\in K}\sum_{i\in I} V_{kint} \qquad \forall n \in N,\ t \in T \tag{1}$$

$$\overline{V^{\mathrm{e}}}_{rt} = \sum_{k\in K} V^{\mathrm{e}}_{krt} \qquad \forall r \in R,\ t \in T \tag{2}$$

$$\overline{V}_{knt} = \sum_{i\in I} V_{kint} \qquad \forall k \in K,\ n \in N,\ t \in T \tag{3}$$

$$\widehat{X}_{gikt} = \sum_{h\in H} X_{gihkt} \qquad \forall g \in G,\ i \in I,\ k \in K,\ t \in T \tag{4}$$

$$\overline{L}_{ghktr} = \sum_{i\in I} L_{gihktr} \qquad \forall g \in G,\ h \in H,\ k \in K,\ t \in T,\ r \in R(h) \tag{5}$$

The first two carry the method. The inventory block reads $\overline{V}_{nt}$ and $\overline{V^{\mathrm{e}}}_{rt}$ and nothing else about the trucks — not which truck carried what, nor how the load was placed.

4.2 Objective

Transport cost counts *dispatches*, so a configured but undispatched truck costs nothing; holding cost falls on erected RTIs.

$$\min\ z^K = \sum_{g\in G}\sum_{k\in K}\sum_{t\in T} cK\cdot Y_{gkt}, \qquad \min\ z^I = \sum_{g\in G}\sum_{n\in N}\sum_{t\in T} ic_{gn}\cdot I_{gnt} \tag{6}$$

$$\min\ \eta^K z^K + \eta^I z^I, \qquad \eta^K = \eta^I = 1 \text{ throughout}, \tag{7}$$

4.3 Inventory, flow balance and coverage

Arrivals are delayed; inventory itself accumulates one period at a time. For the plant,

$$I_{pnt} = I_{p,n,t-1} - d_{pnt} + \overline{V}_{n,t-lt} \qquad \forall n \in N,\ t \geq lt+1 \tag{8}$$

$$I_{pnt} = I_{p,n,t-1} - d_{pnt} + a_{nt} \qquad \forall n \in N,\ t \in \{2,\dots,lt\} \tag{9}$$

$$I_{pnt} = i_{pn} - d_{pnt} + a_{nt} \qquad \forall n \in N,\ t = 1 \tag{10}$$

$$Ie_{ptr} = Ie_{p,t-1,r} + \sum_{n\in M(r)} d_{pnt} - \overline{V^{\mathrm{e}}}_{rt} \qquad \forall r \in R,\ t \geq 2 \tag{11}$$

$$Ie_{ptr} = ie_{pr} + \sum_{n\in M(r)} d_{pnt} - \overline{V^{\mathrm{e}}}_{rt} \qquad \forall r \in R,\ t = 1 \tag{12}$$

Consuming a full RTI produces a folded one in the same period, which is why d_{pnt} drives (11). For the supplier,

$$I_{snt} = I_{s,n,t-1} + d_{snt} - \overline{V}_{nt} \qquad \forall n \in N,\ t \geq 2 \tag{13}$$

$$I_{snt} = i_{sn} + d_{snt} - \overline{V}_{nt} \qquad \forall n \in N,\ t = 1 \tag{14}$$

$$Ie_{str} = Ie_{s,t-1,r} + \overline{V^{\mathrm{e}}}_{r,t-lt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R,\ t \geq lt + 1 \tag{15}$$

$$Ie_{str} = Ie_{s,t-1,r} + ae_{rt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R,\ t \in \{2, \ldots, lt\} \tag{16}$$

$$Ie_{str} = ie_{sr} + ae_{rt} - \sum_{n \in M(r)} d_{snt} \qquad \forall r \in R,\ t = 1 \tag{17}$$

The lead time therefore acts on one side of a balance only — the sending node's stock falls in the dispatch period, the receiving node's rises lt periods later — an alignment Section 7 uses twice. Coverage stock is enforced at both nodes and on both kinds of stock,

$$I_{gnt} \ \geq \ ss_{gnt} \qquad \forall g \in G,\ n \in N,\ t \in T \tag{18}$$

$$Ie_{gtr} \ \geq \ sse_{gtr} \qquad \forall g \in G,\ r \in R,\ t \in T \tag{19}$$

the requirement being that node's own series over the next SC periods:

$$ss_{gnt} = \sum_{j=1}^{SC} d_{g,n,\tau(t,j)}, \qquad sse_{gtr} = \sum_{n \in M(r)} ss_{gnt}, \qquad \tau(t,j) = \begin{cases} t + j, & t + j \leq |T|, \\ |T| - SC + j, & \text{otherwise.} \end{cases} \tag{20}$$

Near the end of the horizon (20) wraps the window back onto the last SC periods rather than truncating it, following (Díaz-Madroñero, Peidro, and Mula 2014): truncating lowers the requirement of the final periods and with it the optimal value, so the tail rule belongs to the problem statement.

4.4 The layout of a truck

Everything from here to (43) is written for one truck: one direction g, one index k, one period t. At most one stack type occupies a slot and only a configured truck has slots; a stack is oriented along one axis or the other; its height cannot exceed the trailer's, which with $r_{gr} > 0$ also empties an unused slot:

$$\widehat{X}_{gikt} \ \leq \ Z_{gkt} \qquad \forall i \in I \tag{21}$$

$$LX_{gihkt} + LY_{gihkt} \ = \ X_{gihkt} \qquad \forall h \in H,\ i \in I \tag{22}$$

$$\sum_{r \in R(h)} r_{gr}\, L_{gihktr} \ \leq \ rK \cdot X_{gihkt} \qquad \forall h \in H,\ i \in I \tag{23}$$

Slots must not overlap — for each pair $i < j$, the usual four-way disjunction closed by a cover row — and every slot lies inside the floor:

$$PX_{gikt} + \sum_{h \in H} \left(p_h LX_{gihkt} + q_h LY_{gihkt}\right) \leq PX_{gjkt} + M^x(1 - A_{gijkt}) \tag{24}$$

$$PX_{gjkt} + \sum_{h \in H} \left(p_h LX_{gjhkt} + q_h LY_{gjhkt}\right) \leq PX_{gikt} + M^x(1 - B_{gijkt}) \tag{25}$$

$$PY_{gikt} + \sum_{h \in H} \left(q_h LX_{gihkt} + p_h LY_{gihkt}\right) \leq PY_{gjkt} + M^y(1 - C_{gijkt}) \tag{26}$$

$$PY_{gjkt} + \sum_{h \in H} \left(q_h LX_{gjhkt} + p_h LY_{gjhkt}\right) \leq PY_{gikt} + M^y(1 - D_{gijkt}) \tag{27}$$

$$A_{gijkt} + B_{gijkt} + C_{gijkt} + D_{gijkt} \ \geq \ \widehat{X}_{gikt} + \widehat{X}_{gjkt} - 1 \tag{28}$$

$$PX_{gikt} + \sum_{h \in H} \left(p_h LX_{gihkt} + q_h LY_{gihkt}\right) \leq pK \tag{29}$$

$$PY_{gikt} + \sum_{h \in H} \left(q_h LX_{gihkt} + p_h LY_{gihkt}\right) \leq qK \tag{30}$$

The containment rows are unconditional, which is what lets the non-overlap rows take the tightened $M^x = pK$, $M^y = qK$.

4.5 Full truckload on every dispatched truck

A dispatched truck must be full, and it can be full in two ways: it may run out of space or of payload. A volume-only floor is not a conservative simplification but an infeasible one — filling the fraction $\alpha = 0.55$ of a 97.92 m^3 trailer ($13\,600 \times 2\,400 \times 3\,000$ mm) with freight at 579 kg/m^3 weighs 31.2 t against a 24 t cap, and seven of the industrial catalogue's fifteen types are denser still. Summing the two ratios is the opposite error: it accepts a truck 35% full by volume and 20% by weight. We therefore require a fraction α of *either* resource, a binary θ_{gkt} selecting which one binds, on the load the truck carries with each commodity's own weight:

$$\sum_{n\in N} \gamma_{p\rho(n)}\, \overline{V}_{knt} \;\geq\; \alpha\, \Gamma_K \big(Y_{pkt} - 1 + \theta_{pkt}\big) \qquad \forall k \in K,\ t \in T \tag{31}$$

$$\sum_{n\in N} w_n\, \overline{V}_{knt} \;\geq\; \alpha\, pK^w \big(Y_{pkt} - \theta_{pkt}\big) \qquad \forall k \in K,\ t \in T \tag{32}$$

$$\sum_{r\in R} \gamma_{sr}\, V^{\mathrm{e}}_{krt} \;\geq\; \alpha\, \Gamma_K \big(Y_{skt} - 1 + \theta_{skt}\big) \qquad \forall k \in K,\ t \in T \tag{33}$$

$$\sum_{r\in R} w^{\mathrm{f}}_r\, V^{\mathrm{e}}_{krt} \;\geq\; \alpha\, pK^w \big(Y_{skt} - \theta_{skt}\big) \qquad \forall k \in K,\ t \in T \tag{34}$$

Exactly one of each pair is active at $Y_{gkt} = 1$, so each is the disjunction "full by volume *or* full by payload". The same disjunction on the layer *capacity* the stacks provide rather than on the load they receive,

$$\sum_{h\in H}\sum_{r\in R(h)} \gamma_{gr}\, n_{hr}\, \overline{L}_{ghktr} \;\geq\; \alpha\, \Gamma_K \big(Y_{gkt} - 1 + \theta_{gkt}\big), \quad \sum_{h\in H}\sum_{r\in R(h)} \hat{w}_{gr}\, n_{hr}\, \overline{L}_{ghktr} \;\geq\; \alpha\, pK^w \big(Y_{gkt} - \theta_{gkt}\big) \tag{35}$$

is a valid tightening, kept because it lets a solver establish a layout before choosing a product assignment. Which form is the *requirement* matters. (31)–(34) is, full truckload being a property of the freight that moves; (35) follows from it because a slot never receives more RTIs of type r than its layers hold and $\hat{w}_{gr} \geq w_n$ for every $n \in M(r)$, and taken *alone* would admit trucks that are not full. That inequality is also why $\hat{w}_{pr} = \max_{n\in M(r)} w_n$ is the family *maximum*: the family *minimum* would assert that *every* product assignment of this layout fills the truck, cutting off feasible loads.

4.6 Dispatch, loading and weight

A dispatched truck must be configured,

$$Y_{gkt} \;\leq\; Z_{gkt} \qquad \forall g \in G,\ k \in K,\ t \in T, \tag{36}$$

while the converse fails: $Z_{gkt} = 1$ with $Y_{gkt} = 0$ is a truck carrying a layout and no freight, which in the reformulated block is exactly the empty column (Section 6).

Full RTIs in a slot cannot exceed the layer capacity of the stack placed there, products sharing an RTI type partitioning it; folded RTIs are capped by the truck's aggregate capacity, folded weight being deterministic in the type. Both loads are gated on the dispatch, at their own analytic bounds rather than a generic big-M:

$$\sum_{n\in M(r)} V_{kint} \;\leq\; \sum_{h\,:\,r\in R(h)} n_{hr}\, L_{p,i,h,k,t,r} \qquad \forall k \in K,\ i \in I,\ r \in R,\ t \in T \tag{37}$$

$$V^{\mathrm{e}}_{krt} \;\leq\; \sum_{h\,:\,r\in R(h)} n_{hr}\, \overline{L}_{s,h,k,t,r} \qquad \forall k \in K,\ r \in R,\ t \in T \tag{38}$$

$$V_{kint} \leq \bar{u}^V_n\, Y_{pkt} \qquad \forall k \in K,\ i \in I,\ n \in N,\ t \in T \tag{39}$$

$$V^{\mathrm{e}}_{krt} \leq \bar{u}^{\mathrm{e}}_r\, Y_{skt} \qquad \forall k \in K,\ r \in R,\ t \in T \tag{40}$$

The payload is a property of the truck, the crush limit of each stack; on plant-bound trucks that cap is *period-exact*, evaluated on the mix actually loaded in the slot, on supplier-bound trucks deterministic in the

layout:

$$\sum_{n\in N} w_n\,\overline{V}_{knt} \;\le\; pK^{w}\,Y_{pkt}, \qquad \sum_{r\in R} w^{\mathrm{f}}_r\,V^{\mathrm{e}}_{krt} \;\le\; pK^{w}\,Y_{skt} \qquad \forall k\in K,\ t\in T \tag{41}$$

$$\sum_{n\in N} w_n\,V_{kint} \;\le\; W^{\max} \qquad \forall k\in K,\ i\in I,\ t\in T \tag{42}$$

$$\sum_{h\in H}\sum_{r\in R(h)} w^{\mathrm{f}}_r\,n_{hr}\,L_{s,i,h,k,t,r} \;\le\; W^{\max} \qquad \forall i\in I,\ k\in K,\ t\in T \tag{43}$$

where (41) names both payload rows and (39) is tighter in the relaxation than their implication, since what bounds a slot's content is the crush cap.

4.7 Symmetry, slot-set sizing and data assumptions

The slots of a truck are interchangeable, and so are the trucks of a direction and period. Both are ordered:

$$\widehat{X}_{g,i+1,k,t} \;\le\; \widehat{X}_{gikt} \qquad \forall g\in G,\ k\in K,\ t\in T,\ i<|I| \tag{44}$$

$$Y_{g,k+1,t} \;\le\; Y_{gkt} \qquad \forall g\in G,\ t\in T,\ k<|K| \tag{45}$$

Re-indexing makes (45) valid, but it removes only *which* subset of trucks is used: loads and layouts stay interchangeable among those used, so a plan dispatching m trucks in a period still appears $m!$ times — the difficulty Section 5 removes by aggregation rather than by a row. The slot count is a modelling parameter, and

$$|I| \;\le\; \Big\lfloor (pK\cdot qK)\big/\min_{h\in H} p_h q_h \Big\rfloor \tag{46}$$

eliminates provably unused slot variables without cutting any feasible layout. It is more than preprocessing: a placement model given fewer slots than a layout needs can report infeasible while the load is placeable (Section 6).

Four properties of the data are assumed: ρ is a partition; $lt \ge 1$, which the lagged terms require; all weights and heights are strictly positive, which makes $\bar{u}^{V}_n$ and $\bar{u}^{\mathrm{e}}_r$ well defined and lets (23) empty an unused slot; and the footprint geometry is consistent, one layer of stack type h tiling its footprint, $n_{hr}a_r = p_h q_h$, with the volume charged to an RTI its bounding box, $\gamma_{gr} = a_r r_{gr}$. *The last is a departure*, and Proposition 2 and the floor-area inequality (72) are both false without it. It costs nothing here, both instance families building one stack type per distinct RTI footprint, so $n_{hr} = 1$ and the relation is an identity.

4.8 Domains

$$Z_{gkt},\, Y_{gkt},\, \theta_{gkt},\, X_{gihkt},\, LX_{gihkt},\, LY_{gihkt},\, A_{gijkt},\, B_{gijkt},\, C_{gijkt},\, D_{gijkt} \in \{0,1\},$$
$$L_{gihktr},\, V_{kint},\, V^{\mathrm{e}}_{krt},\, I_{gnt},\, Ie_{gtr} \in \mathbb{Z}_{\ge 0}, \qquad PX_{gikt},\, PY_{gikt} \ge 0.$$

5 Decomposition and master problem

3D-SCPTOP carries a complete two-dimensional packing problem for every truck of every direction and period — $2|K||T|$ replicas, each with $4\binom{|I|}{2}$ big-M disjunctions (24)–(27) whose linear relaxation is weak — and the replicas are interchangeable, so every solution appears a factorial number of times. Both difficulties disappear when the unit of decision is a complete loaded truck: the geometry moves into one pricing problem per direction, and identical trucks are counted rather than indexed. The Dantzig–Wolfe reformulation below (Dantzig and Wolfe 1960; Barnhart et al. 1998; Desaulniers, Desrosiers, and Solomon 2005) does both without restricting anything (Figure 2; Table 9 in Appendix B maps the rows).

Every row from (21) to (45) carries a single direction, truck and period, and none couples two of them except (45), which relates truck $k+1$ to truck k; the inventory and coverage rows (8)–(19) see the trucks only through the aggregates (1)–(2) and the dispatch count. The formulation is therefore block-angular over $\mathcal{B} = G\times K\times T$, and removing the quantifier over $\mathcal{B}$ from the block rows leaves a mixed integer programme over one truck, the pricing problem (PP^{g}) of Section 6; for the stack-height row,

$$\underbrace{\sum_{r\in R(h)} r_{gr}L_{gihktr} \le rK\,X_{gihkt} \quad \forall g,h,i,k,t}_{\text{3D-SCPTOP, (23)}} \;\longmapsto\; \underbrace{\sum_{r\in R(h)} r_{gr}L_{ihr} \le rK\,X_{ih} \quad \forall h,i,}_{\text{(PP}^{g}\text{), (63)}} \tag{47}$$

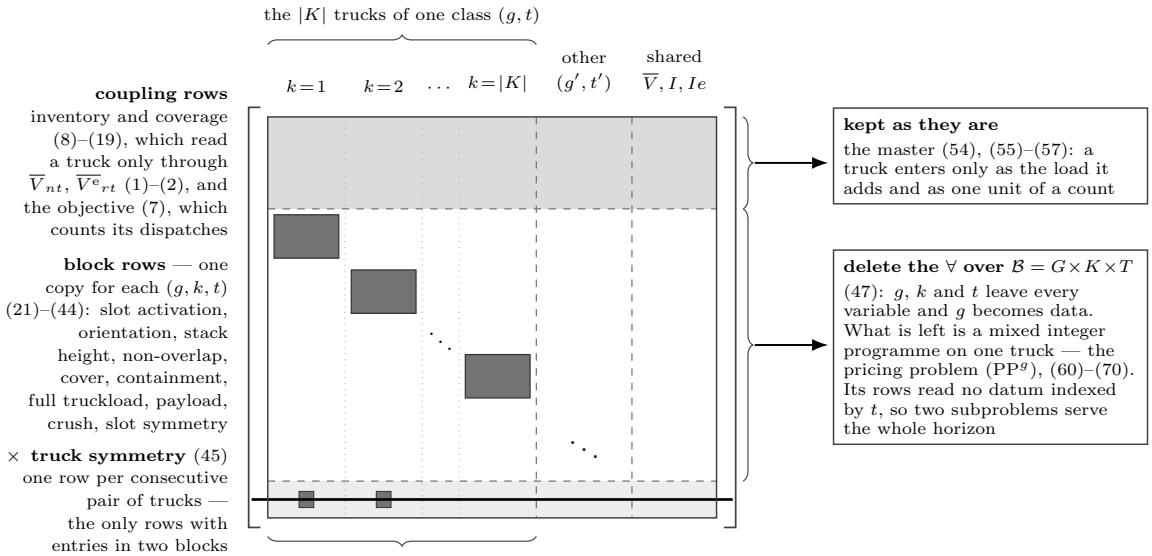


Figure 2: The decomposition read off the formulation. Every geometric, stacking, crush, payload and full-truckload row of Section 4 is confined to one truck, so the matrix is block-angular over $\mathcal{B} = G \times K \times T$; deleting the quantifier over $\mathcal{B}$ from those rows leaves the pricing problem, and no geometry stays behind in the master.

The trucks of a class (g,t) are anonymous, so their blocks are aggregated into one integer variable per block solution, counting the trucks that take it and bounded by $|K|$; the order of the trucks, and with it (45), disappears, and the two binaries of (36) collapse into one (Section 6). The pricing rows read no datum indexed by t, so one pricing problem per direction serves every period, the periods differing only in their duals. Every block variable that reaches the master is integer, so a column may be taken to be an integer point of its pricing problem completed by placement coordinates.

5.1 Columns

A column of a direction is the image of a feasible point of that direction's pricing problem, kept together with the point as a witness; Ω^p and Ω^s are the two column sets, indexed by ω. A *plant-bound* column $\omega \in \Omega^p$ is a vector $b_\omega = (b_{\omega n})_{n \in N}$ of full RTIs for which there exist a slot count $m \le |I|$, stack types $h_1, \dots, h_m \in H$, orientations, bottom-left coordinates (x_i, y_i), layer counts ℓ_{ir} and per-slot product loads v_{in} such that

$$\text{the } m \text{ footprints, rotated as chosen, do not overlap and lie inside } pK \times qK, \tag{48}$$

$$\textstyle\sum_{r \in R(h_i)} r_{pr}\ell_{ir} \;\le\; rK \qquad \forall i \le m, \tag{49}$$

$$\textstyle\sum_{n \in M(r)} v_{in} \;\le\; n_{h_i r}\ell_{ir} \qquad \forall i \le m,\ r \in R, \tag{50}$$

$$\textstyle\sum_{n \in N} w_n v_{in} \;\le\; W^{\max} \qquad \forall i \le m, \tag{51}$$

$$\textstyle\sum_{n \in N} w_n b_{\omega n} \;\le\; pK^w, \qquad \max\Big\{\frac{\sum_n \gamma_{p\rho(n)} b_{\omega n}}{\Gamma_K},\ \frac{\sum_n w_n b_{\omega n}}{pK^w}\Big\} \;\ge\; \alpha, \tag{52}$$

$$b_{\omega n} = \textstyle\sum_{i \le m} v_{in} \qquad \forall n \in N. \tag{53}$$

These are the block rows of Section 4 read on one truck; in (50) $n_{h_i r}\ell_{ir}$ is read as 0 for $r \notin R(h_i)$, and a partly filled layer is a legal layout. A *supplier-bound* column $e_\omega = (e_{\omega r})_{r \in R}$ is the analogous object over folded RTIs, with (43) as its crush condition and $e_{\omega r} \le \sum_{i \le m} n_{h_i r}\ell_{ir}$ (38) in place of (50) and (53). We write (48)–(53) for either direction and $\text{image}_c(\omega)$ for the entry of b_ω or e_ω at the *commodity* c — a product when $g = \text{p}$, an RTI type when $g = \text{s}$ — with w_c, γ_{gc} and r_{gc} its weight, volume and height: w_n, $\gamma_{p\rho(n)}$ and $r_{p\rho(n)}$ for a product, w^{f}_r, γ_{sr} and r_{sr} for an RTI type. Every column costs $c_\omega = \eta^K cK$, and Ω^g holds the

non-empty columns only. Because a column holds its per-slot assignment, (51) is the period-exact cap (42) rather than a worst-case surrogate, and because it keeps its placement, a plan is a set of drawings.

5.2 The master

Let $\lambda^g_{\omega t} \in \mathbb{Z}_{\geq 0}$ be the number of trucks of class (g,t) dispatched carrying column ω. The master problem (MP) is

$$\min \sum_{g\in G}\sum_{t\in T}\sum_{\omega\in\Omega^g} c_\omega\,\lambda^g_{\omega t} \;+\; \eta^I \sum_{g\in G}\sum_{n\in N}\sum_{t\in T} ic_{gn} I_{gnt} \tag{54}$$

subject to

$$\overline{V}_{nt} = \sum_{\omega\in\Omega^p} b_{\omega n}\,\lambda^p_{\omega t} \qquad \forall n\in N,\ t\in T \tag{55}$$

$$\overline{V^{\mathrm{e}}}_{rt} = \sum_{\omega\in\Omega^s} e_{\omega r}\,\lambda^s_{\omega t} \qquad \forall r\in R,\ t\in T \tag{56}$$

$$\sum_{\omega\in\Omega^g} \lambda^g_{\omega t} \leq |K| \qquad \forall g\in G,\ t\in T \tag{57}$$

(8)–(19) verbatim

$$\lambda^g_{\omega t}\in\mathbb{Z}_{\geq 0},\quad I_{gnt},\, Ie_{gtr}\in\mathbb{Z}_{\geq 0},\quad \overline{V}_{nt},\,\overline{V^{\mathrm{e}}}_{rt}\in\mathbb{Z}_{\geq 0}\ \text{(implied by (55)–(56))}.$$

Every column costs the same, so the transport term is cK times the number dispatched, and (55)–(56) are (1)–(2) with each truck's load replaced by its column's image; $n_{gt} = \sum_\omega \lambda^g_{\omega t}$ is the number dispatched by class (g,t), on which Section 7 branches.

Proposition 1 (equivalence)**.** *MP and 3D-SCPTOP have the same optimal value.*

Proof. Each used truck (g,k,t) of a 3D-SCPTOP solution induces a load vector satisfying its direction's column conditions, read off its own layout with $\ell_{ir} = L_{g,i,h_i,k,t,r}$ and, at the plant, $v_{in} = V_{kint}$; every one of those conditions is a row the truck already satisfies. Counting trucks by load vector gives a feasible MP solution of equal cost, since transport cost depends only on the number of used trucks and the inventory rows see only the aggregates (55)–(56) reproduce. Conversely, give each truck of an MP solution the layout witnessing its column: the blocks are identical, so any assignment serves, and in index order the result also satisfies (45). □

No hypothesis is needed, because there is no catalogue of layouts to be sized: a truck's layout is its own.

5.3 The restricted master

MP is solved over a subset $\bar{\Omega}^g$ with integrality relaxed, the restricted master problem (RMP), whose duals the pricing problems use to look for columns of negative reduced cost. With π_{ct} (π_{nt} or π_{rt}) the duals of (55)–(56) and $\mu_{gt}\leq 0$ that of (57), the reduced cost of a plant-bound column dispatched in t is

$$\bar{c}(\omega,t) \;=\; c_\omega - \sum_{n\in N}\pi_{nt}\,b_{\omega n} - \mu_{pt}, \tag{58}$$

and analogously at the supplier; with $z^\star_{gt}$ the optimal value of the pricing problem of class (g,t), $\min_\omega \bar{c}(\omega,t) = z^\star_{gt} - \mu_{gt}$, and a column enters when that is negative. The prices are rounded *upwards* before entering the integer-coefficient subproblem, so that its value over-estimates the true maximum, as the certificate below requires. The RMP is *elastic*: artificials at a large price on (55)–(56), the coverage rows and the cuts keep it feasible for any pool, and a solution with positive artificials is not a plan. It carries cumulative cover cuts: with β^g_{ct} the cumulative requirement of commodity c at the receiving node through t — demand to date plus the coverage stock of t, net of initial stock and freight in transit, floored at zero — and $\nu^g = \min\{\lfloor pK^w/\min_c w_c\rfloor,\ \lfloor \Gamma_K/\min_c \gamma_{gc}\rfloor\}$ the most units one truck can carry, every $t > lt$ gives

$$\sum_{\tau\leq t-lt}\sum_{\omega\in\Omega^g}\lambda^g_{\omega\tau} \;\geq\; \max\left\{\left\lceil\frac{\sum_c \gamma_{gc}\beta^g_{ct}}{\Gamma_K}\right\rceil,\ \left\lceil\frac{\sum_c w_c\beta^g_{ct}}{pK^w}\right\rceil,\ \left\lceil\frac{\sum_c \beta^g_{ct}}{\nu^g}\right\rceil\right\}. \tag{59}$$

Each ratio divides a requirement by the most one truck carries of a resource. These rows, like every branching row of Section 7, are *robust*: their coefficients are constant within a class or affine in the column's image, so their duals fold into μ_{gt} or π_{ct}, (58) keeps its form and no branching decision reaches a pricing problem. A column priced at t receives a dispatch variable at t alone.

5.4 Lower bounds

The RMP is a restriction of MP, so until column generation converges its value is an *upper* bound on the master LP, usable only as a filter. A lower bound needs a certificate: when the pricing problems have *proved*

$$\zeta_{gt} \;:=\; \max_{\omega\in\Omega^g} \sum_c \pi_{ct}\,\mathrm{image}_c(\omega) \;\le\; \eta^K cK - \mu_{gt} \qquad \forall g\in G,\ t\in T,$$

nothing outside $\bar\Omega^g$ prices out, the duals are feasible for the dual of MP, and weak duality gives $z_{\mathrm{RMP}} = z_{\mathrm{MP}} \le z^\star_{\text{3D-SCPTOP}}$ (Algorithm 1), with z_{RMP} and z_{MP} the LP values of the RMP and of MP and $z^\star_{\text{3D-SCPTOP}}$ the optimum. Every pair must be priced to proven optimality with every branching fix released; a call stopped on its time limit forfeits the certificate, while one stopped by the objective cutoff without returning a point proves that no column beats the threshold. The bound is the raw LP value, penalty included, and positive artificials at the end of a proved sweep mean the region admits no plan; an LP the solver does not solve to optimality gives no verdict. A complete sweep that still finds columns yields only the Lagrangian bound $z_{\mathrm{RMP}} + \sum_{g,t} |K| \min(0, \underline{c}_{gt})$, with the artificials at zero and $\underline{c}_{gt} = \eta^K cK - \mu_{gt} - \bar\zeta_{gt}$ formed from an upper bound $\bar\zeta_{gt} \ge \zeta_{gt}$ on the pricing maximum, never from a load that was found; it is too weak here to rely on.

Algorithm 1 CERTIFY$(\bar\Omega, \mathcal{R})$ — the sweep a lower bound rests on, inside a region $\mathcal{R}$: bounds on the fleet vector (Section 7), none at the root. No memoisation, no restriction to the pairs a branch has touched, every branching fix released.

release every branching fix; impose $\mathcal{R}$ on the RMP as fleet rows
loop
  solve the LP over $\bar\Omega$ inside $\mathcal{R}$
  if it is not solved to optimality **then**
    return $(0,\ \text{false})$ ▷ no verdict: the pool may be unable to serve $\mathcal{R}$
  end if
  if the deadline has passed **then**
    return (raw LP value, false)
  end if
  proved ← true
  for all $(g,t) \in G \times T$ **do**
    $\zeta_{gt} \leftarrow$ PRICE(π, μ, g, t) to proven optimality, duals rounded *up*
    if the call returned neither OPTIMAL nor INFEASIBLE, and was not an objective-limit stop with *no* point returned **then**
      proved ← false ▷ a cutoff that returns nothing is still a proof
    end if
  end for
  if some ζ_{gt} prices out **then**
    add its columns to $\bar\Omega$ and **continue**
  end if
  if artificials > 0 **then**
    return $(\infty,\ \text{proved})$ ▷ if proved, no plan over Ω^g lies in $\mathcal{R}$
  end if
  return (raw LP value, proved) ▷ a bound only if proved
end loop

A geometric sweep costs $2|T|$ calls at the pricing limit, which is rarely affordable; every 3D lower bound in this paper comes instead from the volumetric relaxation, by the following inclusion.

Proposition 2 (bound by inclusion)**.** *Let $\Omega^g_{\mathrm{vol}} \supseteq \Omega^g$ be the loads that fit the trailer's volume and the payload and clear the full-truckload floor on the binding resource, and z_{vol} the certified LP value of the master* (54)–(57) *over them. Then $z_{\mathrm{vol}} \le z^\star_{\text{3D-SCPTOP}}$, and the same holds inside any region defined by bounds on the fleet vector.*

Proof. The inclusion rests on two conventions of Section 4: $\Gamma_K = pK \cdot qK \cdot rK$, and the volume γ_{gr} charged to an RTI is its bounding box, with no allowance for void space inside it or clearance between stacks, so

that by footprint consistency one layer of stack type h occupies exactly $n_{hr}\gamma_{gr} = p_h q_h r_{gr}$. A placement satisfying (48)–(53) puts its RTIs in disjoint boxes inside the trailer — footprints non-overlapping and inside $pK \times qK$, each stack under rK by (49) — so their bounding-box volumes sum to at most Γ_K, its weight is at most pK^w, and (52) is the same floor in both sets; hence $\Omega^g \subseteq \Omega^g_{\mathrm{vol}}$. Only the column set depends on the load model, so the 3D master is the volumetric master with columns removed: its LP value is at least z_{vol}, and its optimum at least its LP value. □

Over the volumetric loads the same sweep is a two-constraint knapsack oracle and closes in seconds, so the method asks geometric pricing only for columns and takes its bound from the volumetric relaxation.

6 The pricing problem

The pricing problem of class (g,t) *is* the block of 3D-SCPTOP with the quantifier over $\mathcal{B}$ removed: it designs one truck — its slots, stacks, orientations, positions and load — so as to maximise the dual value of its load, and it is the only place where the geometry lives. The rows are those of Section 4, with g supplying the data and the master the duals; the only additions are the valid inequalities (72). There is one such problem per direction, (PP^g). With c the commodity index and w_c, γ_{gc}, r_{gc} as in Section 5.1, image$_c$ is $\overline{V}_n$ respectively V^{e}_r, and the partition programme (82)–(86) uses the same convention. Figure 3 draws what the block designs.

$$\min\ \eta^K cK \cdot Z\ -\ \sum_c \pi_{ct}\,\mathrm{image}_c \qquad \bar{c} = z - \mu_{gt} \tag{60}$$

$$\widehat{X}_i \le Z \qquad \forall i \in \mathcal{I} \tag{61}$$

$$LX_{ih} + LY_{ih} = X_{ih} \qquad \forall h \in H,\ i \in \mathcal{I} \tag{62}$$

$$\sum_{r \in R(h)} r_{gr}\, L_{ihr} \le rK \cdot X_{ih} \qquad \forall h \in H,\ i \in \mathcal{I} \tag{63}$$

$$PX_i + \delta^x_i \le PX_j + M^x(1 - A_{ij}), \quad PX_j + \delta^x_j \le PX_i + M^x(1 - B_{ij}) \qquad \forall i < j \tag{64}$$

$$PY_i + \delta^y_i \le PY_j + M^y(1 - C_{ij}), \quad PY_j + \delta^y_j \le PY_i + M^y(1 - D_{ij}) \qquad \forall i < j \tag{65}$$

$$A_{ij} + B_{ij} + C_{ij} + D_{ij} \ge \widehat{X}_i + \widehat{X}_j - 1 \qquad \forall i < j \tag{66}$$

$$PX_i + \delta^x_i \le pK, \qquad PY_i + \delta^y_i \le qK \qquad \forall i \in \mathcal{I} \tag{67}$$

$$\widehat{X}_{i+1} \le \widehat{X}_i \qquad \forall i \le |\mathcal{I}| - 1 \tag{68}$$

$$\sum_c \gamma_{gc}\,\mathrm{image}_c\ \ge\ \alpha\,\Gamma_K\,(Z - 1 + \theta) \tag{69}$$

$$\sum_c w_c\,\mathrm{image}_c\ \ge\ \alpha\,pK^w\,(Z - \theta) \tag{70}$$

with $\delta^x_i = \sum_h (p_h LX_{ih} + q_h LY_{ih})$ and $\delta^y_i = \sum_h (q_h LX_{ih} + p_h LY_{ih})$ the footprint of slot i in its chosen orientation, and (67) naming both containment rows and (64), (65) the four non-overlap rows, so that at $Z = 1$ the pair (69)–(70) is the disjunction (52). The truck binaries Z and Y of the compact model differ only at a truck with a layout and no freight, whose image and cost are those of the empty column; the block keeps Z, charges it $\eta^K cK \cdot Z$, and excludes the empty column by fixing $Z = 1$. Four implied rows are available — two restating the floor on the layer *capacity* rather than on the load, and two bounding the occupied floor area — and the oracles carry the second pair:

$$\sum_{h,\, r \in R(h)} \gamma_{gr}\, n_{hr}\, \overline{L}_{hr}\ \ge\ \alpha\,\Gamma_K\,(Z - 1 + \theta), \qquad \sum_{h,\, r \in R(h)} \hat{w}_{gr}\, n_{hr}\, \overline{L}_{hr}\ \ge\ \alpha\,pK^w\,(Z - \theta), \tag{71}$$

$$\sum_i \sum_h p_h q_h X_{ih}\ \le\ pK \cdot qK, \qquad \sum_i \sum_h p_h q_h X_{ih}\ \ge\ \alpha\,pK \cdot qK\,(Z - 1 + \theta). \tag{72}$$

(71) is (35) with the block index dropped, and (72) holds because, under the footprint consistency of Section 4.7, a stack holds at most rK times its footprint area of volume, so a truck full by volume occupies at least a fraction α of its floor.

6.1 The direction-specific rows

Plant-bound, $g = \mathrm{p}$, with $\text{image}_n = \overline{V}_n$:

$$\sum_{n \in M(r)} V_{in} \;\le\; \sum_{h \,:\, r \in R(h)} n_{hr}\, L_{ihr} \qquad \forall i \in \mathcal{I},\, r \in R \tag{73}$$

$$V_{in} \;\le\; \bar{u}^{V}_{n}\, Z \qquad \forall i \in \mathcal{I},\, n \in N \tag{74}$$

$$\sum_{n \in N} w_n\, \overline{V}_n \;\le\; pK^{w}\, Z \tag{75}$$

$$\sum_{n \in N} w_n\, V_{in} \;\le\; W^{\max} \qquad \forall i \in \mathcal{I} \tag{76}$$

and supplier-bound, $g = \mathrm{s}$, with $\text{image}_r = V^{\mathrm{e}}_r$:

$$V^{\mathrm{e}}_r \;\le\; \sum_{h \,:\, r \in R(h)} n_{hr}\, \overline{L}_{hr} \qquad \forall r \in R \tag{77}$$

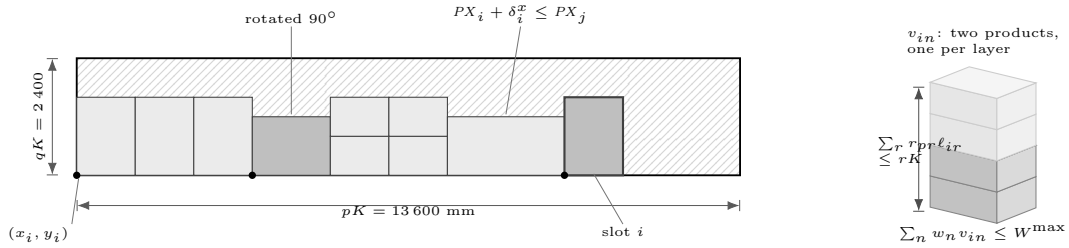
$$V^{\mathrm{e}}_r \;\le\; \bar{u}^{\mathrm{e}}_r\, Z \qquad \forall r \in R \tag{78}$$

$$\sum_{r \in R} w^{\mathrm{f}}_r\, V^{\mathrm{e}}_r \;\le\; pK^{w}\, Z \tag{79}$$

$$\sum_{h} \sum_{r \in R(h)} w^{\mathrm{f}}_r\, n_{hr}\, L_{ihr} \;\le\; W^{\max} \qquad \forall i \in \mathcal{I} \tag{80}$$

The crush limit on a plant-bound truck depends on which products ride in the slot, so it needs the per-slot assignment V_{in}; a supplier-bound truck does not.

The oracle. The oracle must return an optimal point of this feasible set for the duals it is handed, over the true slot bound (46), with the floor imposed on the load carried and any per-commodity cap on the image honoured (Section 7.1). A sweep that only adds columns passes each pair's threshold to the oracle as an objective limit, so a call that returns no point proves that the pair has no improving column; a certifying sweep prices every pair to proven optimality (Algorithm 1). Over the volumetric loads of Appendix A — volume and payload within the trailer's, floor cleared on the binding resource — pricing is a two-constraint integer knapsack, and the same master serves both load models.



(a) the floor, (64)–(67) (b) one slot, (49) and (51)

Figure 3: What the pricing problem designs: footprints placed on the floor without overlap, each slot a stack of homogeneous layers under the height budget (63) and the per-slot crush cap, the load assigned to the slots, $b_{\omega n} = \sum_{i \le m} v_{in}$, and the truck full by volume *or* by payload (69)–(70).

What pricing cannot return. The pricing objective is linear in the image, so its optimum is an extreme point of $\mathrm{conv}(\Omega^g)$. A balanced load inside the hull is dominated by a fractional blend of single-commodity trucks, and no dual vector returns it. Such interior loads are what an integer plan needs when a period's demand does not decompose into full trucks; the construction of Section 7.2 builds them directly, and branching on the fleet moves the duals so that pricing is asked for loads the pool lacks.

7 Branching, with the flow-first construction inside it

The search branches on quantities the master already prices, so every child is a set of robust rows and the pricing problem is never changed. Transport costs $cK \cdot X$ with $X = \sum_{g,t} n_{gt}$ the total fleet, and moving

freight between periods can compensate a bound on a single period's dispatches or deliveries at the price of one period's holding cost but not a bound on X; the search therefore branches on X first and then, inside $X = X^\star$, on the fleet vector. Bounding a single $\lambda^g_{\omega t}$ instead would exclude a column from the subproblem (Vanderbeck 2000), and with many columns differing in a single RTI the relaxation would only shift the same fractional weight one column over. Master, pricing problem, construction and tree together form BP-3D-SCPTOP (Algorithm 2).

Level 1: the total fleet. Around an incumbent of total $X^\star$, the regions $X \geq X^\star + 1$, $X = X^\star$ and $X \leq X^\star - 1$ partition the space. The first costs a whole truck and is usually pruned by bound; the second fixes transport, so everything in it is inventory allocation; the third is the only one that can beat the incumbent outright, and it is expanded first. A plan found inside an inequality region splits it the same way, so level 1 recurses.

Level 2: the fleet vector, inside $X = X^\star$. Fix an order on the pairs (g, t). Any plan in the region differs from the incumbent's fleet vector $\mathbf{n}^\star$ at a first index j, or nowhere, so branching there gives the disjoint and exhaustive children

$$n_j \leq n^\star_j - 1, \qquad n_j = n^\star_j, \qquad n_j \geq n^\star_j + 1, \tag{81}$$

with the equality child partitioned again at $j + 1$ (Murty 1968) and expanded first, always around the plan constructed at the node.

Pruning, bounds and what may be claimed. A node is a region imposed on the RMP as fleet rows. Its pool LP is a filter; where the filter allows and the policy admits the node, the region is certified (Algorithm 1) and pruned if the certified value reaches the incumbent, and otherwise it keeps its parent's bound. The policy certifies the root, the regions $X \leq X^\star - 1$, a region that can be split no further, and, in a reserve at the end, the frontier; a Murty chain stops at depth 6, raised by 2 whenever the stack empties within the budget (Table 10). An infeasible node relaxation is not a proof that a region is empty, stock non-negativity being rigid; only a proved sweep that leaves artificials positive is, or arithmetic: the fleet box, availability caps telescoped from the sending rows, and the constants of (59) read as a requirement floor. Under 3D loading every region's certificate comes from the volumetric relaxation with the region's fleet rows applied, which Proposition 2 covers. The incumbent is the cheapest verified plan any construction found and the global bound the minimum over the frontier, a dropped region contributing the root's bound. Optimality is claimed only when the root is certified, no region remains open or merely bounded, no region was dropped, and the incumbent meets the global bound; no run here meets those conditions. A third level, on deliveries, is disabled in every run reported, so a full-depth leaf is covered by a certificate rather than split again.

7.1 The initial pool

The search starts from one maximally full truck per commodity and direction, obtained by pricing that commodity alone with every other capped at zero (volumetrically, by filling to capacity); no mixed loads are seeded.

7.2 The flow-first construction

The primal heuristic decides how much moves and how many trucks carry it before it builds any truck. Pricing returns only extreme loads, so the mixed loads a plan needs must come from elsewhere: the construction fixes integer deliveries and fleet over *continuous* column weights, then cuts each period's deliveries into legal trucks. It runs inside the region $\mathcal{R}$ of its node, on a fresh copy of the RMP (Algorithm 3).

1. *Restricted master.* The LP over the pool, inside $\mathcal{R}$, gives the duals and the fractional point.
2. *Aggregate integer master.* The dispatch totals n_{gt} become integer variables and $\overline{V}_{nt}, \overline{V^{\mathrm{e}}}_{rt}$ integer, the column variables stay continuous, $\mathcal{R}$ is imposed on the n_{gt}, and under 3D loading the aggregate forms of (87)–(89) are added. The objective is decided here, without a pricing call; artificial slack above a tolerance abandons the node.

Algorithm 2 BP-3D-SCPTOP: branch and bound over the fleet vector, with CONSTRUCT as primal heuristic and CERTIFY as bound.

$\bar{\Omega} \leftarrow$ one maximally full single-commodity truck per commodity and direction, each verified
$\mathrm{LB}_0 \leftarrow \textsc{Certify}(\bar{\Omega}, \mathrm{root})$; $(\bar{z}, \mathbf{n}^\star, \bar{\Omega}) \leftarrow \textsc{Construct}(\bar{\Omega}, \mathrm{root})$; $X^\star \leftarrow \sum_{g,t} n^\star_{gt}$
$\mathcal{S} \leftarrow [\, X \geq X^\star + 1,\ X = X^\star,\ X \leq X^\star - 1 \,]$ ▷ a stack: the region that can beat the incumbent is popped first
while $\mathcal{S} \neq \emptyset$ and time remains **do**
    $\mathcal{R} \leftarrow$ pop $\mathcal{S}$; impose it on the RMP as fleet rows
    if $\mathcal{R}$ is refuted arithmetically **then**
        continue ▷ fleet box, availability, requirement
    end if
    $\tilde{z} \leftarrow$ LP over $\bar{\Omega}$ inside $\mathcal{R}$ ▷ over-estimates the region's LP: a filter, never a bound
    if $\tilde{z} \geq \bar{z}$ and the policy certifies $\mathcal{R}$ **then**
        $(\mathrm{LB}_{\mathcal{R}}, \mathrm{proved}) \leftarrow \textsc{Certify}(\bar{\Omega}, \mathcal{R})$
        if $\mathrm{LB}_{\mathcal{R}} \geq \bar{z}$ **then**
            continue ▷ pruned
        end if
        record $\mathrm{LB}_{\mathcal{R}}$ as an open bound
    else
        $\mathcal{R}$ carries its parent's bound
    end if
    $(z, \hat{\mathbf{n}}, \bar{\Omega}) \leftarrow \textsc{Construct}(\bar{\Omega}, \mathcal{R})$ ▷ its columns join the shared pool whatever it returns
    if no plan **then**
        price under $\mathcal{R}$'s duals and construct once more; **continue** if still none
    end if
    if $z < \bar{z}$ **then**
        $(\bar{z}, \mathbf{n}^\star) \leftarrow (z, \hat{\mathbf{n}})$ ▷ new incumbent
    end if
    if $\mathcal{R}$ does not fix X **then**
        push the three regions of level 1 around $\sum_{g,t} \hat{n}_{gt}$ ▷ level 1 recurses
    else if the Murty depth j is below the cap (Table 10) **then**
        push the three children (81) at the j-th pair, around $\hat{\mathbf{n}}$, equality last
    end if
end while
for all $\mathcal{R}$ still on $\mathcal{S}$, while a reserve of time remains **do**
    filter by the pool LP; certify the survivors; record their bounds ▷ the frontier
end for
$\mathrm{LB} \leftarrow \max\{\mathrm{LB}_0, \min$ over the open and frontier bounds$\}$, capped at $\bar{z}$ ▷ a dropped region contributes LB_0
verify every dispatched truck of the incumbent against (48)–(53)
return $(\bar{z}, \mathrm{LB})$; optimality is claimed only if LB_0 was proved, $\mathcal{S} = \emptyset$, no region was dropped, no restart occurred and $\bar{z} = \mathrm{LB}$

3. *Partition, as a cover.* Each period's integer deliveries $Q^\star_{gt}$ are covered by $n^\star_{gt}$ legal trucks, an integer

programme over per-truck commodity counts a_{kc}:

$$\min \ \sum_c \gamma_{gc}\Big(\sum_k a_{kc} - Q^\star_{gtc}\Big) \tag{82}$$

$$\text{s.t.} \ \sum_k a_{kc} \ \geq \ Q^\star_{gtc} \qquad \forall c \tag{83}$$

$$\textstyle\sum_c \gamma_{gc} a_{kc} \leq \Gamma_K, \qquad \sum_c w_c a_{kc} \leq pK^w \qquad \forall k \tag{84}$$

$$\textstyle\sum_c \gamma_{gc} a_{kc} \geq \alpha\Gamma_K \theta_k, \qquad \sum_c w_c a_{kc} \geq \alpha pK^w(1-\theta_k) \qquad \forall k \tag{85}$$

$$\textstyle\sum_c w_c a_{kc} \ \geq \ \sum_c w_c a_{k+1,c} \qquad \forall k < n^\star_{gt} \tag{86}$$

$$a_{kc} \in \{0, \dots, \bar{a}_{gc}\}, \qquad \theta_k \in \{0,1\}.$$

with θ_k selecting the resource on which part k is full and $\bar{a}_{gc}$ the placeability cap below. If $n^\star_{gt}$ parts cannot be formed, $n^\star_{gt}+1$ and $+2$ are tried.

4. *Integer master over pool plus parts*, warm-started with the partition; its solution is the node's plan.

The partition must be a cover: a fractional blend of legal loads need not split into legal loads for any number of trucks, floor area being consumed in whole footprints. A cover is sound because every part is a legal column, step 4 may dispatch fewer of them than were built, and over-delivery is feasible.

7.3 The construction under 3D loading

Under 3D loading each part must also satisfy (48)–(53). The partition programme is a relaxation of placeability, so each part is submitted to a placeability oracle; everything rests on placeability being *downward closed*: removing units from a layout leaves a layout.

Deciding placeability. A constructive *shelf design*, a small integer programme laying the part's stacks in shelves along the trailer, supplies most positive answers, and its layout passes the independent verifier of Section 8.1. Refutations come from decoupling: a stack carries one footprint class, so a load within the payload is placeable if and only if, for each class f, the least number $\underline{s}_f$ of class-f stacks its units need can be placed on the floor as rectangles. The least count is a small cutting-stock problem per class, and the placement is decided exactly by a position-indexed model over the normal patterns of the floor (Christofides and Whitlock 1977; Beasley 1985); an infeasible placement at a lower bound on $\underline{s}_f$ already refutes.

Cuts. A part *proved* unplaceable yields a *dominance cut* forbidding every load that dominates it, valid by downward closure and written at the smallest point $\lfloor \sigma a \rfloor$ of its ray, $\sigma \in (0,1]$, still proved unplaceable. A part left *undecided* is excluded as a single point: a restriction of the heuristic, never a validity claim. The largest placeable count of each commodity is bisected once per run and handed to the partition as $\bar{a}_{gc}$.

Necessary conditions of placeability as rows. Per truck k and footprint class f — a stack type h of Section 4, with $p_f = p_h$ and $q_f = q_h$, and C_f the commodities whose RTI types it carries — the partition carries an integer number of stacks $s_{kf} \leq \lfloor pK \cdot qK / p_f q_f \rfloor$ with the crush limit, the height budget and the floor area imposed:

$$\sum_{c \in C_f} w_c a_{kc} \leq W^{\max} s_{kf}, \qquad \sum_{c \in C_f} r_{gc}\, a_{kc} \leq rK\, s_{kf}, \qquad \sum_f p_f q_f\, s_{kf} \leq pK \cdot qK, \tag{87}$$

where a class carries one RTI type the height row takes the height-limited count, $\sum_{c \in C_f} a_{kc} \leq \lfloor rK / r_{gr} \rfloor s_{kf}$. It also carries the finer conditions by which the oracle refutes a load. With $L^{\max}_f$ the most units of them one stack can hold, a two-resource knapsack under rK and $W^{\max}$,

$$\sum_{\substack{c \in C_f \\ w_c \geq \xi}} a_{kc} \leq \Big\lfloor \frac{W^{\max}}{\xi} \Big\rfloor s_{kf} \ \ \forall \xi \in \{w_c\}_{c \in C_f}, \qquad \sum_{\substack{c \in C_f \\ r_{gc} \geq \xi}} a_{kc} \leq \Big\lfloor \frac{rK}{\xi} \Big\rfloor s_{kf} \ \ \forall \xi \in \{r_{gc}\}_{c \in C_f},$$
$$\sum_{c \in C_f} a_{kc} \leq L^{\max}_f\, s_{kf}, \tag{88}$$

the height thresholds being written only where a class carries more than one RTI type; and, with $s_{kf} = s^0_{kf} + s^1_{kf}$ split into stacks laid with p_f along the trailer's length pK and along its width qK,

$$\sum_f \big(u_1(p_f)\, q_f\, s^0_{kf} + u_1(q_f)\, p_f\, s^1_{kf}\big) \leq U_1\, qK, \qquad \sum_f \big(p_f\, u_2(q_f)\, s^0_{kf} + q_f\, u_2(p_f)\, s^1_{kf}\big) \leq pK\, U_2, \tag{89}$$

Algorithm 3 CONSTRUCT($\bar{\Omega}, \mathcal{R}$) — the flow-first relax and fix. Line 4, the rows of line 12 and lines 15–27 apply under 3D loading only. ε_2 is step 2's relative MIP gap and *tol* the tolerance above which artificial slack means the region admits no plan over this pool.

1: $\bar{\Omega}' \leftarrow$ a fresh copy of the RMP over $\bar{\Omega}$ ▷ not the node's own LP: step 2 changes variable types
2: solve LP($\bar{\Omega}', \mathcal{R}$) ▷ duals π, μ and the fractional point the node stands on
3: make $n_{gt}, \overline{V}_{nt}, \overline{V^{\mathrm{e}}}_{rt}$ integer; keep λ continuous; impose $\mathcal{R}$ on n_{gt} and on λ
4: add the aggregate forms of (87)–(89) ▷ 3D only; layer coefficient = the height count, the crush limit being its own row
5: solve to relative gap ε_2
6: **if** no solution, or artificials $>$ tol **then**
7: **return** $\perp$ ▷ node abandoned; the region is infeasible over this pool
8: **end if**
9: $(Q^\star, n^\star) \leftarrow$ the integer deliveries and fleet
10: **for all** $(g,t) \in G \times T$ **do**
11: **for** $m = n^\star_{gt}$, $n^\star_{gt}+1$, $n^\star_{gt}+2$ while $\mathcal{R}$ allows **do**
12: solve PARTITION($Q^\star_{gt}, m$), (82)–(86), with $\bar{a}_{gc}$ as variable bounds and, in 3D, (87)–(89) per part
13: **break** if feasible ▷ a timeout is not a proof of infeasibility
14: **end for**
15: **for all** parts a_k of the cover **do**
16: (verdict, layout) $\leftarrow$ PACK(a_k)
17: **if** placeable **then**
18: verify the layout; $\bar{\Omega} \leftarrow \bar{\Omega} \cup \{(a_k, \text{layout})\}$
19: **end if**
20: **if** *proved* unplaceable **then**
21: bisect $\sigma \in (0,1]$ and cut every load dominating $\lfloor \sigma a_k \rfloor$ ▷ validity
22: **end if**
23: **if** undecided **then**
24: exclude a_k as a single point ▷ heuristic restriction, never validity
25: **end if**
26: **end for**
27: re-solve the partition if a cut was written and rounds remain
28: **end for**
29: $\bar{\Omega} \leftarrow \bar{\Omega} \cup \{\text{the parts}\}$; solve the integer master over $\bar{\Omega}$, warm-started with the partition
30: **return** $(z, \hat{\mathbf{n}}, \bar{\Omega})$ ▷ the parts are kept even when no plan is returned

for every pair (u_1, U_1) on pK and (u_2, U_2) on qK of a finite family of *dual-feasible functions*, (u, U) being one on a length C when $\sum_i x_i \le C$ implies $\sum_i u(x_i) \le U$ (Fekete and Schepers 2001; Clautiaux, Alves, and Valério de Carvalho 2010): the identity ($U = C$, the area row), the thresholds $[x \ge \xi]$ at the footprint lengths ξ that occur ($U = \lfloor C/\xi \rfloor$), and the scalings $\lfloor jx/C \rfloor$ ($U = j$, $j \le 12$). Every row holds for every placeable truck with s_{kf} and its split read off the truck's own layout — a stack bears at most $W^{\max}$ and rK, and (89) is the conservative-scale argument of Fekete and Schepers (2004) with one dimension unscaled — so the rows remove only parts the oracle would refuse. Step 2 carries their sum over the period's trucks, the right-hand sides of (89) multiplied by n_{gt}. They are implied in the pricing problem and in the master, whose points and columns are placeable, but not in the volumetric relaxation that supplies the bounds; in the method they cut only the free integer vectors of steps 2 and 3.

8 Computational results

8.1 Setup

The method is implemented in Julia with JuMP and solves every model with Gurobi; Appendix C describes the implementation and the settings of every run, and the code and data are available at [repository URL].

Every 3D plan below is re-derived from its dispatch list alone and checked truck by truck against (48)–(53) by a routine independent of the search and of the oracle that built the column, re-deriving each slot's weight from the products it carries. A *gap* is always $(z - \mathrm{LB})/z$ against the method's own certified bound; Table 4 defines the other quantities.

Instances. Instances are named $I_|N|_|T|_|R|_v$, with v the number of erected RTIs that fill one truck by volume. Footprints come from the industrial catalogue, from 800×600 to 2400×1200 mm, with heights quantised to 50 mm and bounded so that at least two layers fit under the roof; demand deviates $\pm 10\%$ from its mean, and content weight by $\pm 25\%$ between products sharing a type, which makes the crush limit depend on the period's mix. Every instance here has $|R| = 2$ and $v = 90$. The smallest, $I_8_5_2_90$, consumes roughly 2.5 truckloads per period; the other six pair $(|N|, |T|)$ with 2.5, 5 or 8 truckloads per period, the name's suffix being that figure times ten: $(8, 10)$, $(12, 12)$ and $(16, 15)$ at 2.5, $(16, 15)$ and $(24, 20)$ at 5, and $(24, 20)$ at 8. The five solved in 3D (Table 6), from $I_8_5_2_90$ to $I_24_20_2_90_k50$, form the *size ladder*, each instance a *rung*. The truck is a semi-trailer, $13600 \times 2400 \times 3000$ mm, payload 24 t, $W^{\max} = 1200$ kg, $\alpha = 0.55$, $cK = 1000$ per dispatch, $lt = 1$, one period of coverage stock.

Table 4: What each reported quantity means.

z	Cost (54) of the plan returned, which carries no artificial slack; an upper bound on the optimum.
trucks	$\sum_{g,t} n_{gt}$ for that plan.
LB	For the method, the certified bound of Section 5.4, printed only where every pair of its sweep was proved; for a compact model, the solver's dual bound at its stop.
gap	$(z - \mathrm{LB})/z$ against the method's own bound; an optimality gap only to the extent that bound is tight.
stop	OPTIMAL, TIME_LIMIT or MEM_LIMIT; after a time or memory stop the row's time is the budget.
nodes / open	Tree nodes opened; regions still unexpanded at the deadline. A run with open regions has not searched the space, and one with none has not refuted it either if a region was closed with a bound below the incumbent.
dropped	Regions abandoned without a certificate of their own (for instance, no plan was constructed in them); only the root's bound covers them.
verified	Dispatched trucks found legal by the independent check of Section 8.1, out of the number dispatched.
Δz	The cost of one method relative to another on the same instance, with the denominator named where it appears.

8.2 The volumetric problem

Table 5: The method under its volumetric oracle against SCPTOP, the compact formulation of the volumetric problem of Appendix A: a check on the method where an independent optimum exists. z is the incumbent at the stop; OPT marks a proved optimum, ^{m} a stop on the memory cap, ^{t} a stop on time. "bound" is Gurobi's dual bound for the compact model and the certified bound of Section 5.4 for the method. $\Delta z = (z_{\text{method}} - z_{\text{compact}})/z_{\text{compact}}$, negative when the method's plan is cheaper. Both sides had 3 600 s (1 800 s for the compact model on $I_8_5_2_90$, which it solved in 0.2 s), 20 GB and ten threads. A volumetric column carries no layout, so no plan here is verified truck by truck.

	compact SCPTOP				BP-3D-SCPTOP (volumetric oracle)					
Instance	z	bound	trucks	t (s)	z	bound (cert.)	trucks	t (s)	Δz (%)	tighter bound
$I_8_5_2_90$	18 796.96 OPT	18 796.96	12	0.198	18 796.96^{t}	18 795.56	12	3064	0.0000	18 796.96 (compact)
$I_8_10_2_90_k25$	38 855.65^{m}	38 851.93	26	3269	38 853.17^{t}	38 791.19	26	3062	−0.0064	38 851.93 (compact)
$I_16_15_2_90_k25$	68 139.17 OPT	68 139.17	45	1350	68 139.21^{t}	68 131.48	45	3117	+0.0001	68 139.17 (compact)
$I_16_15_2_90_k50$	134 346.91^{m}	133 417.47	89	1122	134 507.47^{t}	133 484.56	89	3073	+0.1195	133 484.56 (method)
$I_24_20_2_90_k50$	200 266.21^{m}	199 518.04	137	2093	200 048.69^{t}	200 013.57	137	3071	−0.1086	200 013.57 (method)
$I_24_20_2_90_k80$	282 101.77^{m}	280 823.53	189	2403	281 233.28^{t}	280 910.22	188	3133	−0.3079	280 910.22 (method)

The volumetric problem of Appendix A is the only part of the problem with an independent optimum, because a compact formulation of it, SCPTOP, solves exactly on the smaller instances; Table 5 therefore checks whether the method arrives where it should, at the same 3 600 s, 20 GB and ten threads. It does. On $I_8_5_2_90$ it returns the proved optimum 18 796.96, to the cent, with twelve trucks; on $I_16_15_2_90_k25$ it returns 68 139.21 against a proved 68 139.17 at forty-five trucks, 0.0001% above the optimum, with its own certified bound 0.0113% below it. Where the compact model works it is far the faster, at 0.2 s and 1 350 s against the method's roughly 3 070 s.

On the four instances where the compact model exhausts its memory cap, costs agree within 0.31%, with the same fleet on three and one truck fewer for the method on $I_24_20_2_90_k80$; the method's plan is cheaper on three and dearer on one, by 0.1195% on $I_16_15_2_90_k50$, where it left its own interval 0.76% wide. On the three largest instances its certified bound is the tighter, by 67, 496 and 87 cost units. On $I_12_12_2_90_k25$ the method certifies 52 105.31, valid for the 3D problem by Proposition 2, 274 s into its run, where the compact volumetric model stood at a dual bound of 51 944.22 when its 600 s expired: the decomposition's bound is 0.310% tighter, relative to the compact bound.

8.3 The 3D problem

Table 6: BP-3D-SCPTOP on the size ladder. *root* is the plan the construction returns before any node is popped, with the time it arrived, and *tree* the improvement the search then makes on it, relative to the root plan's cost. *open* and *dropped* are as in Table 4; on row 4, four of the seven open regions carry no bound of their own. LB is the certified volumetric bound, valid for 3D by Proposition 2 and loose wherever the crush limit binds, so the gaps on the last two rows are not optimality gaps; on rows 2 and 4 the compact volumetric model supplies tighter valid bounds (38 851.93 and 68 139.17, Table 5). The method proves no row optimal; row 1's plan attains the compact volumetric optimum (18 796.96), which proves it optimal by inclusion. The wall clock stops short of the budget because 15% of it is reserved for the frontier sweep.

Instance	z	trucks	verified	LB (cert.)	gap
$I_8_5_2_90$	**18 796.96**	12	12 / 12	18 795.56	0.0075%
$I_8_10_2_90_k25$	38 854.44	26	26 / 26	38 791.19	0.163%
$I_12_12_2_90_k25$	52 116.76	34	34 / 34	52 105.31	0.022%
$I_16_15_2_90_k25$	79 703.31	55	55 / 55	68 131.48	14.5%
$I_24_20_2_90_k50$	223 944.20	156	156 / 156	200 013.57	10.7%

Instance	root (t)	tree	nodes	dropped	open	wall / budget (s)
$I_8_5_2_90$	18 797.59 (142 s)	0.003%	301	1	6	765 / 900
$I_8_10_2_90_k25$	38 867.74 (154 s)	0.034%	15	2	9	770 / 900
$I_12_12_2_90_k25$	52 215.42 (298 s)	0.19%	24	5	6	1 548 / 1 800
$I_16_15_2_90_k25$	92 251.99 (948 s)	**13.6**%	2	0	7	1 534 / 1 800
$I_24_20_2_90_k50$	223 944.20 (2 772 s)	0%	1	1	2	3 094 / 3 600

Table 6 takes the method to twenty periods and 156 trucks, every plan verified truck by truck and every gap taken against a certified bound. On the first three rungs the truck count is the minimum possible: each run prunes the region with one truck fewer by the requirement floor of Section 7, which is arithmetic over volume, payload and unit count and so valid under 3D loading. What remains of those gaps, 0.0075%–0.163%, is inventory cost. The gaps of 14.5% and 10.7% on the last two rungs are attributed chiefly to the bound, on the mechanism of Section 8.5; nothing here proves that attribution or measures its share, and those two plans are verified upper bounds under a stated budget.

Within one run, the tree improves the root construction by 13.6% on the fourth rung, where two level-1 nodes take a 67-truck root plan at 92 251.99 to 65 trucks and then to 55 at 79 703.31; by 0.19%, 0.034% and 0.003% on the third, second and first; and not at all on the fifth, whose root plan arrived, after one failed construction, under 290 s before the search window closed, enough for one node, in which the region with fewer trucks returned no plan. Where the volumetric relaxation is nearly exact the construction is already close; where it is not, the region with one truck fewer is where the plan improves.

The volumetric certificate takes 1.3 to 13.8 s once root column generation stops, every pair proved; the geometric sweep certified no region on any run. Placeability tests cost 0.2 to 14.2 s per run, and on the

last two rungs the construction takes most of the time (the fourth rung's root construction 669 s, the fifth's two root attempts 919 and 938 s). Of the ladder's 8 335 placeability calls none was left undecided: the shelf design placed 6 749 loads and the exact test placed 485 and proved 1 101 unplaceable, none by the arithmetic conditions alone, which the partition carries as rows.

8.4 The compact 3D model

Table 7: The compact 3D-SCPTOP model (a catalogue of $n_{\mathrm{LC}} = 3$ layouts shared by the trucks of a direction, 1 800 s) against BP-3D-SCPTOP, whose rows are those of Table 6 and whose budget is in the last-but-one column. *none*: no feasible solution within the budget. The last column is the interval containing $z^\star$: the method's plan above, and below the tighter bound *on the problem* — a compact volumetric dual bound where one exists, else the method's certificate. The compact 3D model's own dual bound bounds its a-priori restriction, not $z^\star$, and appears in its own column only.

	compact 3D-SCPTOP, 1 800 s				BP-3D-SCPTOP					
Instance	z	trucks	LB	stop	z	trucks	LB (cert.)	gap	budget (s)	$z^\star$ in
*I*_8_5_2_90	18 816.13	12	18 795.76	TIME	18 796.96	12	18 795.56	0.0075%	900	[18 796.96, 18 796.96]
*I*_8_10_2_90_*k*25	*none*	—	36 458.75	TIME	38 854.44	26	38 791.19	0.163%	900	[38 851.93, 38 854.44]
*I*_12_12_2_90_*k*25	*none*	—	48 516.86	TIME	52 116.76	34	52 105.31	0.022%	1 800	[52 105.31, 52 116.76]
*I*_16_15_2_90_*k*25	*none*	—	64 053.23	TIME	79 703.31	55	68 131.48	14.5%	1 800	[68 139.17, 79 703.31]
*I*_24_20_2_90_*k*50	not run				223 944.20	156	200 013.57	10.7%	3 600	[200 013.57, 223 944.20]

A compact formulation of 3D-SCPTOP, written with a catalogue of $n_{\mathrm{LC}} = 3$ layouts shared by the trucks of a direction — a restriction of Section 4 chosen a priori, so its dual bound bounds the restriction and not $z^\star$ — returns a plan on one of four ladder instances (Table 7). On *I*_8_5_2_90, at 67 674 variables and 92 092 constraints, it stops on its 1 800 s limit with twelve verified loads, 0.102% above the method's plan at twice the method's budget. From ten periods on, at 119 424 to 322 734 variables, it returns no feasible solution in 1 800 s, stopping on time with peak resident sets of 6.1, 2.7 and 4.5 GB. Beyond five periods it supplies neither a plan nor a bound on the problem.

*I*_8_5_2_90 is the one instance with a reported optimum, and the proof is not the method's: the compact volumetric relaxation solves it to proved optimality at 18 796.96 in 0.198 s, and the verified 3D plan attains that value. The method's own run ends with one region dropped, six open and three bounded below the incumbent at 18 795.56, 18 795.54 and 18 795.54. On this instance the crush limit never halves a container family's layer count, as it does on *I*_16_15_2_90_*k*25, where Section 8.5 measures the over-credit that causes.

8.5 Is a volumetric capacity adequate?

What a volumetric capacity costs shows in the volumetric plan itself (Table 8). Every load the compact volumetric plan dispatches is put to the method's placeability oracle: 41 of 45 on *I*_16_15_2_90_*k*25, whose volumetric plan is the proved optimum, and 119 of 137 on *I*_24_20_2_90_*k*50 are proved unplaceable, none left undecided, so a volumetric optimum can be a plan nine tenths of which cannot be loaded.

Nor can the plan be repaired by re-packing its own delivery quantities into loadable trucks. In one period of the first instance and five of the second, the plan ships the period's empty containers in a single truck filled to 97–100% of its volume. A folded container takes a larger share of the truck's volume than of its payload, so a truck of empties is full only at $\alpha = 55\%$ of its volume, and two trucks cannot both be full with that quantity. One truck cannot carry it either: in four of the six periods the quantity exceeds the largest count of that container type that one truck can be proved to hold, and in the other two, which mix two types, the method's step 3 finds no loadable truck for it. Re-packing the rest of the plan exactly leaves those empties unshipped, and the plan then fails the supplier's coverage stock in later periods; delivering more than the plan's quantities, as a cover does, takes from the plant folded containers it needs for its own coverage. A loadable plan has to choose different quantities, which is what BP-3D-SCPTOP does: its verified plans use 55 and 156 trucks.

Deflating the capacity is the obvious remedy: adding the stack-count rows (87) to the compact volumetric model raises its answer on *I*_16_15_2_90_*k*25 from forty-five trucks to forty-eight and its dual bound to 70 435.04 after 600 s, 2 296 above that model's own bound of 68 139.17, and leaves the fleet unchanged where the crush limit does not bind (34 trucks on *I*_12_12_2_90_*k*25 at 600 s). and not sufficient, whether those forty-eight loads can be placed has not been tested.

Table 8: The compact volumetric model's plan put to the method's placeability oracle, on the two instances where the per-stack crush limit binds. On $I_16_15_2_90_k25$ the volumetric plan is a proved optimum; on $I_24_20_2_90_k50$ it is a memory stop. A composition dispatched in several periods counts once among the distinct ones. The last volumetric row counts the periods whose delivery quantities the method's step 3 cannot split into loadable full trucks, all of them shipments of empty containers (see the text), so the volumetric plan has no loadable counterpart with the same quantities. BP-3D-SCPTOP, which chooses its own quantities, is given for reference, every truck verified.

	$I_16_15_2_90_k25$	$I_24_20_2_90_k50$
Compact volumetric plan		
trucks dispatched	45	137
proved unplaceable	**41**	**119**
placed	4	18
undecided	0	0
distinct compositions proved unplaceable	39 of 43	111 of 129
periods not split into loadable full trucks	1	5
BP-3D-SCPTOP		
verified trucks / cost	55 / 79 703.31	156 / 223 944.20

The mechanism, and a test a planner can run. The over-credit has one cause, measured on $I_16_15_2_90_k25$: the payload admits 57 full RTIs of product P_1 in one truck while 32 is the largest number that can be placed, the placeability cap the method bisects to proof. The container is 650 mm tall, so four layers fit under the 3 000 mm roof, while P_1 at 419 kg full fits only two under the 1 200 kg crush limit, as do six of the family's eight products. A volumetric fill therefore credits a truck with roughly twice that family's placeable count, which is also why the certified bound on the two crush-binding rungs sits 14.5% and 10.7% below the method's plan.

The mechanism can be checked before anything is solved. For one container type, let the *analytic credit ratio* κ be the count a volumetric capacity credits over an analytic count of how many can be carried,

$$\kappa_r = \frac{\min\{\Gamma_K/\gamma_{pr},\ pK^w/\bar{w}_r\}}{\min\{s_r \min(L_r^h, L_r^c),\ pK^w/\bar{w}_r\}}, \qquad L_r^h = \left\lfloor \frac{rK}{r_{pr}} \right\rfloor, \quad L_r^c = \left\lfloor \frac{W^{\max}}{\bar{w}_r} \right\rfloor, \tag{90}$$

with $\bar{w}_r$ the catalogued gross weight of one full RTI and s_r the number of its footprints a single orientation tiles onto the floor. The denominator is a grid count, an upper estimate of what fits, so $\kappa > 1$ is a conservative diagnosis of over-credit. Over the fifteen types of the industrial catalogue κ has median 1.000, ranges to 2.000 and exceeds 1.20 on four types, and no type has $\kappa < 1$: the error cannot run in the planner's favour, and a plan over a mixed catalogue fails on its worst container. The instances here are generated, with κ in a range the generator controls, so this is a realism check on that range rather than a claim that their results transfer to a particular plant.

9 Conclusions

Procurement transport planning with returnable transport items has been formulated with transport capacity as what it physically is — a three-dimensional loading condition on every dispatched truck — and solved by branch and price over complete loaded trucks. Because 3D-SCPTOP is written over trucks it is block-angular, so the Dantzig–Wolfe master is obtained by aggregation and carries no loading configuration (Table 9), and the equivalence needs no hypothesis. Every lower bound reported rests on Proposition 2, which makes a bound affordable where a geometric certificate is not. And a volumetric transport capacity is not a conservative simplification: most loads of the two volumetric plans checked cannot be placed, and the plans cannot be repaired by re-packing their own quantities, which in six periods do not split into loadable full trucks (Table 8); on the industrial catalogue the error runs one way only.

Three limits are real, and each names its next step. The method certifies no optimum here: the search ends with regions unrefuted, and the one optimum reported is established by inclusion from an exactly solved relaxation. The bound is loose where the crush limit halves a container family's layer count; the stack-count rows (87) tighten it when added to the compact volumetric relaxation, so they and the rows (88)–(89), which

the volumetric relaxation does not imply, are the candidates for tightening the bound inside the search, once it is known whether the loads they admit can be placed. Closing the regions the search leaves unrefuted may need a branching rule on *pairs* of columns. And the comparison against a procedure in industrial use is to be run on instances a plant solves itself, against a plan supplied to it.

A The SCPTOP benchmark

The volumetric benchmark extends the procurement transport model of (Díaz-Madroñero, Peidro, and Mula 2014) to the closed-loop two-node setting studied here: fuzzy parameters are dropped, and transport capacity is expressed through the volume of the individual RTIs against the truck volume rather than a maximum container count. To isolate the effect of the loading representation it is identical to the transport and inventory block of 3D-SCPTOP in every other respect — same two nodes, closed loop, lead time, coverage stock, payload cap, full-truckload floor and objective — and differs in one place: where 3D-SCPTOP asks whether the RTIs can be stacked and placed on the floor, SCPTOP asks only whether their volumes sum to less than the trailer's. The notation is that of Section 4, with $\overline{V}_{knt}$ a decision variable here rather than the slot sum (3).

$$\text{Min } z = \sum_{g\in G}\sum_{k\in K}\sum_{t\in T} cK\, Y_{gkt} + \sum_{g\in G}\sum_{n\in N}\sum_{t\in T} ic_{gn} I_{gnt} \tag{91}$$

subject to, for all $k \in K$ and $t \in T$,

$$\sum_{n\in N} \gamma_{p\rho(n)} \overline{V}_{knt} \le \Gamma_K\, Y_{pkt}, \qquad \sum_{r\in R} \gamma_{sr} V^{\mathrm{e}}_{krt} \le \Gamma_K\, Y_{skt} \tag{92}$$

$$\sum_{n\in N} w_n \overline{V}_{knt} \le pK^{w} Y_{pkt}, \qquad \sum_{r\in R} w^{\mathrm{f}}_{r} V^{\mathrm{e}}_{krt} \le pK^{w} Y_{skt} \tag{93}$$

together with the full-truckload floor on the binding resource, imposed exactly as in (52),

$$\sum_{n} \gamma_{p\rho(n)} \overline{V}_{knt} \;\ge\; \alpha\Gamma_K \big(Y_{pkt} - 1 + \theta_{pkt}\big), \qquad \sum_{n} w_n \overline{V}_{knt} \;\ge\; \alpha\, pK^{w} \big(Y_{pkt} - \theta_{pkt}\big), \tag{94}$$

with $\theta_{gkt} \in \{0,1\}$ and the supplier direction analogous. The inventory balances, coverage stock and domains are (8)–(19) unchanged, and (45) is carried here too, so neither compact model is held to an easier standard. Constraint (92) is the whole of the benchmark's loading model: no footprint, stack height, layer count or crush limit appears.

B The block rows and the pricing problem

Table 9 maps each block row of Section 4 to the row of (PP^g) it becomes when the quantifier over $\mathcal{B}$ is deleted (Section 5).

Table 9: The block rows of Section 4 and the rows of (PP^g) they become when the $\forall$ over $\mathcal{B} = G \times K \times T$ is deleted. The two directions share the geometry, so it is written once with g as data; the direction-specific rows are those of Section 6.1. One row of (PP^g) is not a transcription and is marked as such.

Row	Section 4	(PP^g)
Block rows: the pricing problem		
Slot active	(21)	(61)
Orientation	(22)	(62)
Stack height	(23)	(63)
Non-overlap	(24)–(27)	(64)–(65)
Cover	(28)	(66)
Containment	(29)–(30)	(67)
Full truckload, volume	(31), (33)	(69)
Full truckload, payload	(32), (34)	(70)
The same on capacity	(35)	(71), re-gated on Z

Row	Section 4	(PP^g)
Occupied floor area	— (no counterpart)	(72), a valid inequality
Slot capacity	(37)	(73)
Folded capacity	(38)	(77)
Dispatch gate, full	(39)	(74)
Dispatch gate, folded	(40)	(78)
Payload, both directions	(41)	(75), (79)
Crush limit, plant-bound	(42)	(76)
Crush limit, supplier-bound	(43)	(80)
Slot symmetry	(44)	(68)
Transport cost	(6)	$\eta^K cK \cdot Z$ in (60)
Dispatch	(36)	deleted; Z carries the truck
Truck symmetry	(45)	dropped in the aggregation
Coupling rows: the master		
Aggregate flows	(1)–(2)	(55)–(56)
Fleet per class	$\|K\|$ trucks per (g, t)	(57)
Inventory and coverage	(8)–(19)	verbatim in the master
Holding cost	(6)	second term of (54)

C Implementation and reproducibility

The method is implemented in Julia with the JuMP modelling layer, and every model it solves — masters, the construction's programmes, pricing and placeability models — is solved with Gurobi 13.0; the compact reference models were solved with Gurobi 12.0. A run follows Algorithm 2: root column generation within a share of the budget, the volumetric certificate, the root construction, then the tree until a reserve in which the open regions are certified. A placeability query tries the shelf design first and the exact test of Section 7.3 second, and its verdict is kept for the run. Every 3D plan is verified truck by truck before it is reported, and every certified bound came back with every pair proved. Each instance was solved once, alone on one workstation; since the phases run on wall-clock budgets, a run depends on the build, the machine and its load, and a difference of one or two trucks between rows is not read as an effect. Table 10 lists the settings; code and data: [repository URL].

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Table 10: Settings of the reported runs.

	3D runs (Tables 6, 7)	volumetric runs (Table 5)
budget (wall clock)	900, 900, 1 800, 1 800, 3 600 s	3 600 s
root column generation	15% of the budget	15% of the budget
frontier reserve	15% of the budget	15% of the budget
initial pool, per pricing call	10 s	—
step 2, per call	300 s at the root, 120 s at a node, gap 10^{-4}	same
step 3, per call	60 s, 2 cut rounds, up to $n^{\star}_{gt} + 2$ trucks	same
step 4, per call	budget/6: 150, 150, 300, 300, 600 s	120 s
rows (87)–(89)	steps 2 and 3	—
pricing, per call	20 s	20 s
placeability, per call	20 s	—
certified regions	root; $X \leq X^{\star} - 1$; regions that cannot be split; the frontier	same
Murty depth cap	6, raised by 2 when the stack empties within the budget	same
threads	10	10
memory cap	20 GB per model; 1 GB on volumetric pricing models	same
seed	solver default	solver default
compact models	Gurobi 12.0, 10 threads, 20 GB, MIP gap 0, seed 0; 1 800 s (3D), 600–3 600 s (volumetric)	
check and re-pack, Table 8	the method's placeability oracle, 60 s per load; its step 3 with the rows on the plan's exact quantities, 600 s per call, up to 40 cut rounds, 20 s per placeability call, trucks per period from the volume and payload minimum up to +12; 10 threads, 20 GB	

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